

A LOW-MEMORY SPECTRAL-CORRELATION ANALYZER FOR DIGITAL
QAM-SRRC WAVEFORMS

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DEDICATION

To my dad, for his unconditional love and support.

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ABSTRACT

Cyclostationary signal processing (CSP) can reveal periodic statistical features in received waveforms without prior synchronization. This thesis investigates an FPGA-targeted VHDL architecture for blind symbol-rate and center-frequency estimation of M -ary quadrature amplitude modulation (QAM) waveforms shaped by a square-root-raised-cosine (SRRC) pulse, with resource-constrained CubeSat cognitive radios as the target platform. The proposed streaming spectral correlation analyzer processes samples one at a time: it mixes each candidate center frequency to baseband, applies a two-tap finite-impulse-response filter, estimates the zero-lag cyclic autocorrelation at a candidate symbol rate, and compares the resulting magnitude-squared statistic with an empirically selected, power-scaled threshold. MATLAB models reproduce the expected cyclic features, and fixed-point VHDL simulations driven by a Python transmitter and channel model characterize modulation order, pulse-shape roll-off, requested search-grid spacing, capture length, word length, noise, sample rate, and two-signal cases. Every one of the thirteen archived single-trial plots contains at least one threshold crossing. Eleven experiments report the transmitted symbol-rate–center-frequency pair or pairs; Experiment VII reports a 1.00-MBd symbol rate for a 0.10-MBd input, and Experiment X reports a 0.04-MHz center frequency for a 0.05-MHz input. The sample-by-sample datapath avoids an FFT sample buffer by construction. Its serial search time grows with the capture length and the number of evaluated candidate pairs, while its effective frequency spacing is limited by the numerically controlled oscillator’s frequency-control-word width. Recovered 2021 component reports and project scripts identify a ZedBoard XC7Z020CLG484-1 and

Vivado 2019.2.1, but no top-level synthesis report links the archived experiment-level resource counts to that setup or documents their BRAM-count convention. A recovered, source-divergent 2022 checkout also contains accumulator, power-format, and flow-control discrepancies, so it cannot independently establish bit-true conformance of the thesis-era simulations to the ideal statistic. These results are simulations and reported resource counts, not flight-hardware measurements.

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CHAPTER I

INTRODUCTION

Many digitally modulated radio-frequency (RF) communication signals exhibit second-order statistical properties that vary periodically with time. This behavior is known as *cyclostationarity*. This thesis focuses on estimating the cycle frequencies of second-order cyclostationary (SOC) signals. In this work, we perform temporal analysis using the cyclic autocorrelation function (CAF) and spectral analysis using the cyclic spectral density function (CSD) [7].

Some standard CSD estimation techniques are the frequency-smoothing method (FSM), time-smoothing method (TSM), strip spectral correlation analyzer (SSCA), and fast Fourier transform (FFT) accumulation method (FAM) [7]. A realization of a CSD estimator is called a *spectral correlation analyzer* (SCA) [3, 7]. This thesis investigates SCA techniques and evaluates their suitability for severely resource-constrained platforms such as CubeSats.

Terrestrial receivers can often accommodate less stringent size, weight, and power (SWaP) constraints than spacecraft receivers. Higher-SWaP devices such as multicore central processing units (CPUs), graphics processing units (GPUs), and large-cell-count field-programmable gate arrays (FPGAs) can therefore be deployed to perform complex digital signal processing (DSP) routines.

For CubeSat cognitive radios (CRs), however, onboard resources are constrained. A candidate CR may need to collect data, process signals, receive commands, report

status, and store data while meeting spacecraft SWaP limits. Resource-conscious devices such as smaller FPGAs are therefore relevant implementation platforms.

Many established SCA algorithms rely extensively on FFTs, whose data buffering can consume substantial block random access memory (BRAM) in FPGA implementations. This thesis studies a time-domain SCA design that relies minimally on BRAM. The low-BRAM design operates on a sample-by-sample (“streaming”) basis and avoids an FFT sample buffer. This makes the architecture a candidate for resource-constrained CubeSat CR research, although the thesis does not establish flight readiness.

Chapter II begins by reviewing the basic notation and concepts of stochastic processes, also known as *random signals*. We then develop the intuition and mathematics behind cyclostationary signal processing (CSP). Next, we discuss the challenges associated with calculating the power spectral density (PSD) of random signals. PSD estimation techniques are discussed, followed by CSD estimation techniques for cyclostationary signals. We then evaluate several SCAs and discuss their suitability for our low-BRAM objective. Finally, we conclude with an overview of a technique that leverages CSP to intelligently reuse spectral bands, known as *spectrum sensing*.

In Chapter III, we first discuss the motivation for performing spectrum sensing in space. Next, we discuss a set of assumptions necessary for the design of our low-BRAM SCA. In MATLAB, we model a CAF-discrete Fourier transform (CAF-DFT) estimator that adapts a previously studied symbol-rate detector and center-frequency sweep to a serial hardware architecture. We demonstrate it using M -ary quadrature amplitude modulation (M-QAM) with square-root-raised-cosine (SRRC) pulse shaping, abbreviated M-QAM-SRRC. With a resource-constrained target in mind, we simplify parts of the design using a single-point estimator with little BRAM. We conclude this chapter with a comprehensive look at the digital architecture for this design, which we refer to as the *streaming SCA* because of its BRAM-reducing,

stream-based approach.

In Chapter IV, we characterize our streaming SCA. The experiments stress-test various parameters, including bit widths, symbol rates, center frequencies, capture lengths, noise, and SRRC roll-off factors. We conclude this thesis in Chapter V with an overall summary and a discussion of opportunities for future work.

CHAPTER II

BACKGROUND

In this chapter, we review the concept of a random signal. We then briefly discuss the autocorrelation function (AF), followed by the CAF. After this, we describe the challenges of PSD and CSD estimation. Finally, we conclude with an overview of a technique that leverages CSP to reuse free spectral bands, known as *spectrum sensing*.

2.1 Cyclostationary Signal Processing

In this section, we summarize random signal notation and concepts as described in W. A. Gardner's book *Introduction to Random Processes with Applications to Signals and Systems* [2]. We begin with the AF and then build on this groundwork to introduce the concept of the CAF.

2.1.1 The Autocorrelation Function

We begin with a review of stochastic processes. For digital communications, these are typically referred to as *random signals*. A random signal $X(t)$ is a collection of random variables indexed by time. One outcome of the underlying random experiment produces a sample path, or time series, denoted by $x(t)$. Communication signals are often represented by complex-valued random processes, whereas the familiar one-dimensional Gaussian density describes a real-valued random variable $D \sim \mathcal{N}(\mu, \sigma^2)$ as follows:

$$p_D(d) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{d-\mu}{\sigma}\right)^2}. \quad (1)$$

The Gaussian PDF has the shape shown in Figure 1.

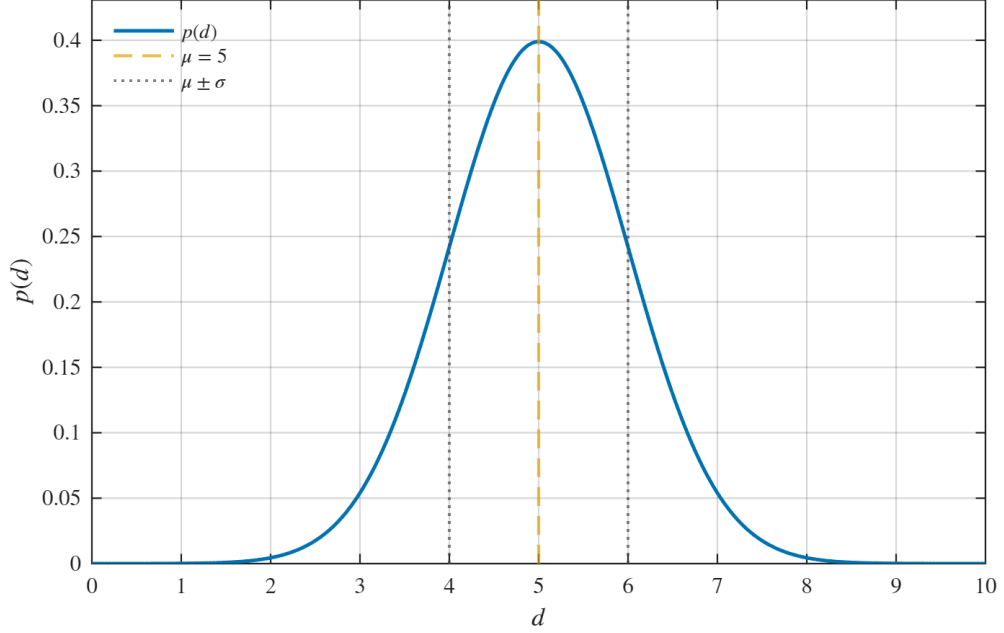


Figure 1: Example of a Gaussian PDF. Here, Equation (1) is evaluated for 500 values of d , with a mean of $\mu = 5$ and a standard deviation of $\sigma = 1$.

A random process is completely specified by all of its finite-dimensional joint distributions, or by the corresponding joint PDFs when those densities exist. Its mean is an ensemble average over realizations at a fixed time. For a real-valued process with marginal PDF $p_{X(t)}(x)$, the mean is

$$\mu_X(t) = E\{X(t)\} = \int_{-\infty}^{\infty} x p_{X(t)}(x) dx. \quad (2)$$

For a complex-valued process, the expectation in Equation (2) is applied to the real and imaginary parts. The ensemble average is distinct from a time average along one sample path; the two are equal only under the relevant ergodicity conditions. The

first raw moment $E\{X(t)\}$ is the mean $\mu_X(t)$ by definition, without any Gaussian or symmetry assumption [2].

The second central moment $\sigma_X^2(t) = E\{|X(t) - \mu_X(t)|^2\}$ is the *variance* of $X(t)$. For a real-valued process, the magnitude squared reduces to the ordinary square. The nonnegative square root $\sigma_X(t)$ is the *standard deviation*, which describes the spread of $X(t)$ about its mean [2].

The second joint moment $R_X(t_1, t_2) = E\{X(t_1)X^*(t_2)\}$ is the *AF*. For WSS random signals, the AF is expressed as $R_X(\tau)$, where $\tau = t_1 - t_2$. The variable τ is the time separation, or *lag*, between the two samples [2].

If the joint distributions of $X(t)$ through order n are invariant under a common shift of all their time arguments, the process is said to be *n*th-order stationary. A process with this invariance at every order is *strict-sense stationary*. A finite-second-moment process is second-order stationary (SOS), or *wide-sense stationary* (WSS), when its mean is constant and its AF is invariant under a common time shift, so that the AF depends only on lag [2, 7].

Let us further develop $X(t)$ by assuming it is a WSS random signal. Its AF is

$$R_X(\tau) = E \left\{ X \left(t + \frac{\tau}{2} \right) X^* \left(t - \frac{\tau}{2} \right) \right\}. \quad (3)$$

Thus, a WSS process has a constant mean and variance, and its AF is a function of τ rather than of absolute time. For a correlation-ergodic WSS process, $R_X(\tau)$ can also be recovered as the infinite-time average along almost any sample path; a finite-record version is an autocorrelation estimate [2]. Figure 2 illustrates a WSS process with constant mean and variance.

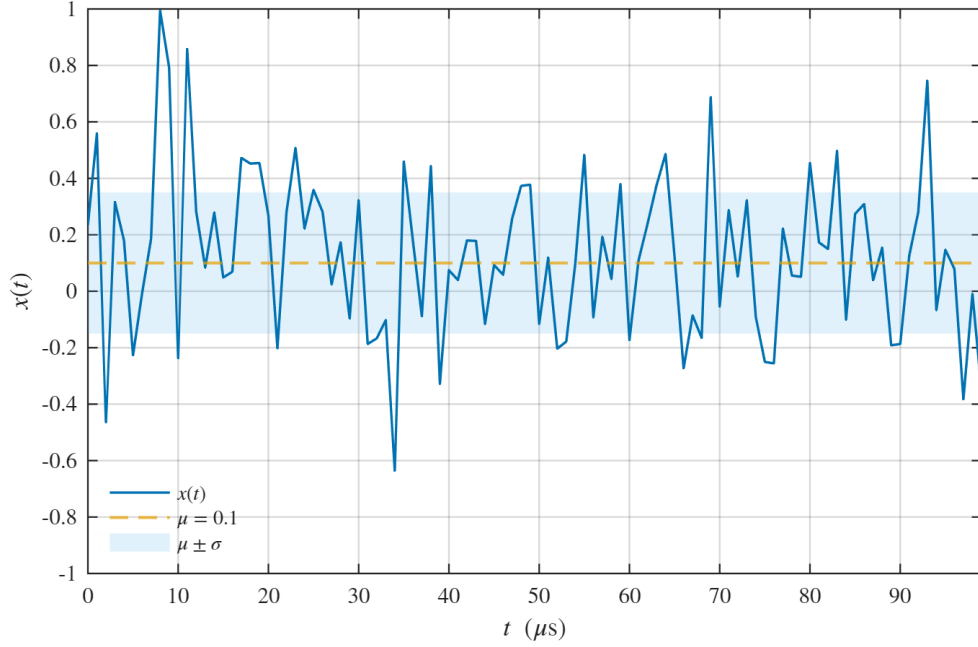


Figure 2: One 100-sample realization of an independent and identically distributed (IID) Gaussian process with constant mean $\mu_X = 0.1$ and standard deviation $\sigma_X = 0.25$. The dashed line is the theoretical mean and the shaded band is $\mu_X \pm \sigma_X$.

Many digitally modulated communication waveforms are not WSS but instead exhibit cyclostationarity. In this context, we use *waveform* to mean a modulated signal. Quadrature amplitude modulation (QAM) conveys information through the amplitudes of two orthogonal carriers at the same frequency, conventionally called the in-phase and quadrature components. Each symbol selects one of M constellation points, where M is the *modulation order*. Under the MATLAB convention used in this thesis, the $M = 2$ case reduces to two antipodal constellation points and is equivalent to binary phase-shift keying (BPSK). The rate at which symbols are transmitted is the *symbol rate* (SR), denoted R_s , and the modulated carrier frequency is the waveform's *center frequency* (CF), denoted f_c . An example of a 2-QAM time series is shown in Figure 3.

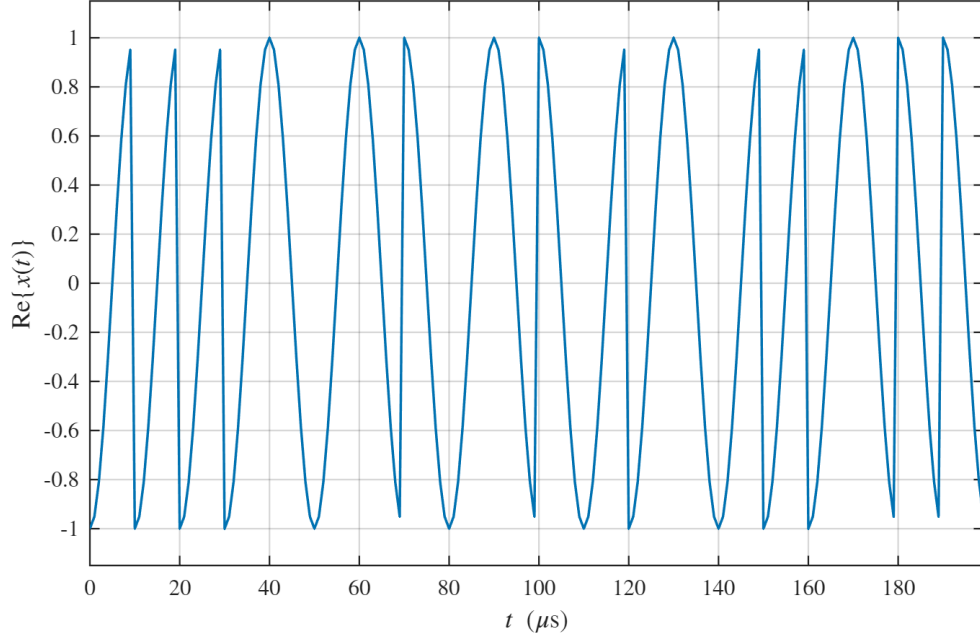


Figure 3: The real component of a 2-QAM waveform. The model generates 400 random symbols, applies a length-10 rectangular pulse, and displays the first 200 samples. The waveform parameters are $R_s = 0.1$ MBd, $f_c = 0.05$ MHz, and $F_s = 1$ MHz.

An M -QAM waveform has the following mathematical representation:

$$X(t) = \sum_{k=-\infty}^{\infty} a_k p(t - kT_0 - t_0) e^{j(2\pi f_c t + \theta_0)}, \quad (4)$$

where a_k is the k th data symbol, $p(t)$ is the pulse shape, $T_0 = 1/R_s$ is the symbol period, t_0 is the timing offset, f_c is the CF, and θ_0 is the carrier-phase offset. This linear-modulation model includes QAM and phase-shift keying (PSK) waveforms. Adaptive modulation and Nyquist pulse shaping are widely used communication techniques. This thesis narrows its investigation to the M-QAM-SRRC class of waveforms.

To conclude this section, Figure 4 plots $|\hat{R}_x(\tau)|$ versus τ in what is known as an *autocorrelogram*.

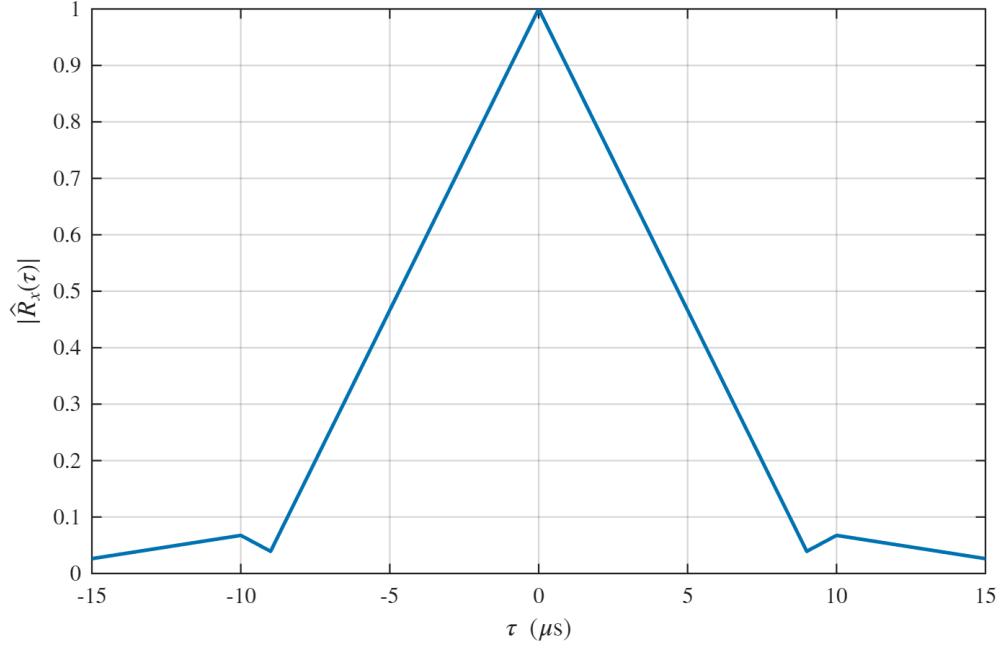


Figure 4: Magnitude of the biased autocorrelation estimate for the waveform $x(t)$ shown in Figure 3, evaluated at 31 integer-sample lags.

2.1.2 The Cyclic Autocorrelation Function

If the finite-dimensional distributions of $X(t)$ are periodic under a common shift of their time arguments, $X(t)$ is said to exhibit *cyclostationarity*. If its mean and AF are periodic under a common time shift, it is said to be *SOC* or *wide-sense cyclostationary* (WSC) [2, 7]. The ensemble CAF is the Fourier-series coefficient of the symmetric time-varying AF $R_X(t, \tau) = E\{X(t + \tau/2)X^*(t - \tau/2)\}$:

$$\tilde{R}_X^\alpha(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} R_X(t, \tau) e^{-j2\pi\alpha t} dt. \quad (5)$$

In Equation (5), α is the *cycle frequency*. More generally, the cyclic expectation $E^\alpha\{Z(t)\}$ denotes the α -frequency Fourier coefficient of the time-varying ensemble mean $E\{Z(t)\}$; it is not a different ordinary expectation. Under the appropriate correlation-cycloergodicity condition, the ensemble CAF equals the fraction-of-time

average computed from a single sample path [7]. The resulting sample-path expression is

$$R_X^\alpha(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x\left(t + \frac{\tau}{2}\right) x^*\left(t - \frac{\tau}{2}\right) e^{-j2\pi\alpha t} dt. \quad (6)$$

Figure 5 plots the magnitude of a finite-sample estimate of Equation (6) in what is known as the cyclic autocorrelogram.

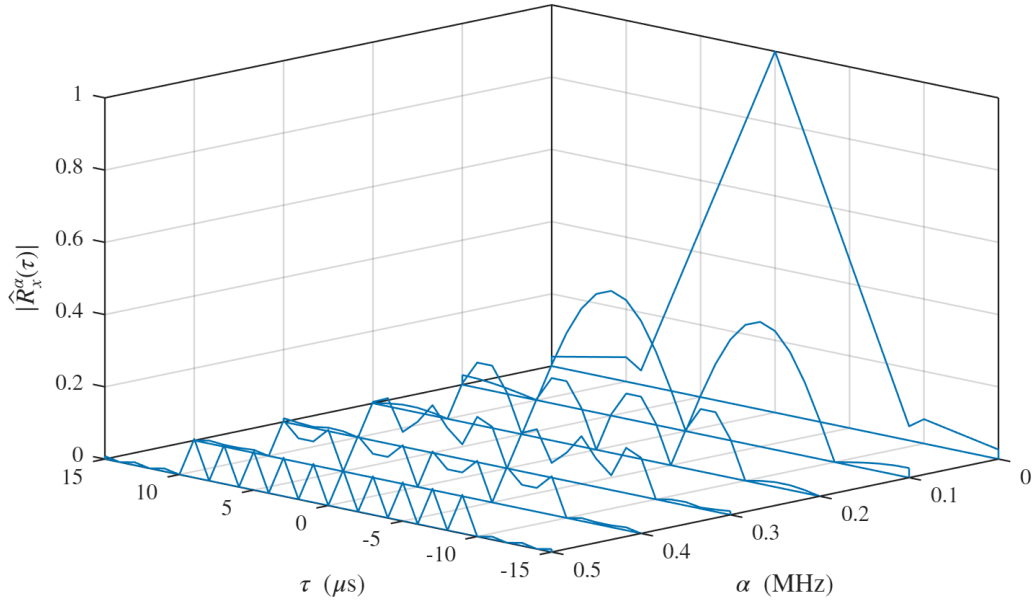


Figure 5: Magnitude of the biased cyclic-autocorrelation estimate for the waveform $x(t)$ shown in Figure 3, evaluated at 31 integer-sample lags and six cycle frequencies from 0 through 0.5 MHz.

Peaks in the finite-sample cyclic autocorrelogram indicate candidate cycle features of $X(t)$, providing information not present in the ordinary AF alone. We will build on this concept in the following three sections to explain the significance of cyclic features.

2.2 Spectral Density

This section continues the summary of notation and concepts described in [2]. We first review the periodogram, the challenges faced when estimating the PSD from the periodogram, and a PSD estimation method known as *frequency-smoothing*. We then build upon this work to develop the concept of the CSD.

2.2.1 The Power Spectral Density Function

A rectangular pulse is defined as

$$\Pi(t) = \begin{cases} 0, & \text{if } |t| > \frac{1}{2} \\ 1, & \text{if } |t| \leq \frac{1}{2} \end{cases}. \quad (7)$$

The Fourier transform of $\Pi(t)$ is denoted $\hat{\Pi}(f)$. Since $\Pi(t)$ is a finite-energy signal, its energy spectral density is defined by

$$\Psi_{\Pi}(f) = \left| \hat{\Pi}(f) \right|^2. \quad (8)$$

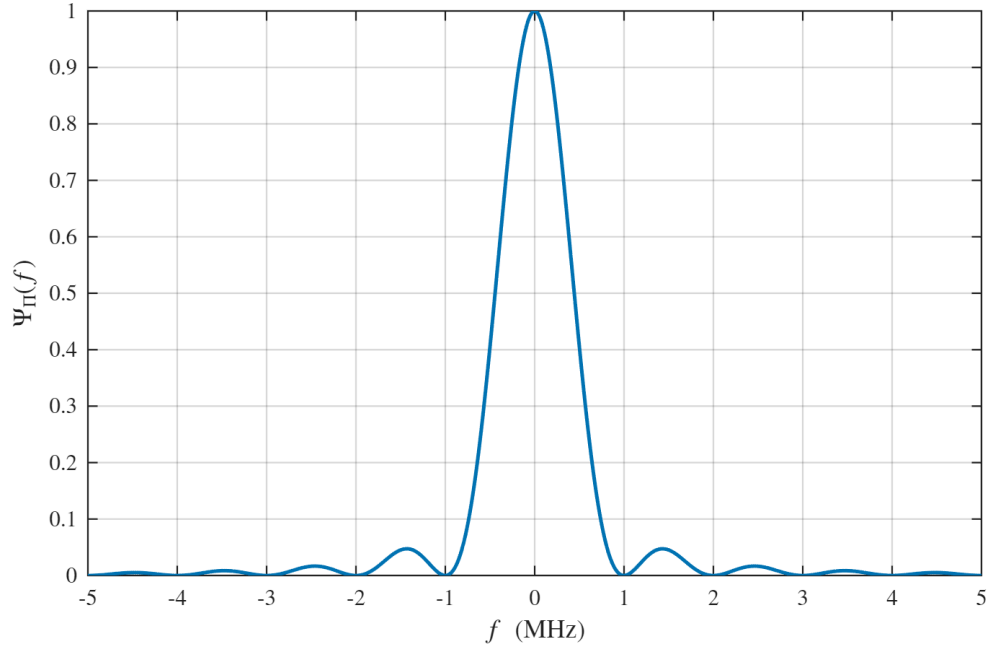


Figure 6: Squared-magnitude Fourier transform of the unit-width rectangular pulse $\Pi(t)$. At $F_s = 50$ MHz, the one-microsecond pulse occupies 50 samples within a 150-sample record; a zero-padded 4096-point frequency grid renders the curve smoothly, and the numerical ESD satisfies Parseval’s energy equality.

Parseval’s theorem states that the total spectral energy, $\int_{-\infty}^{\infty} \Psi_{\Pi}(f) df$, equals the time-domain energy, $\int_{-\infty}^{\infty} |\Pi(t)|^2 dt$ [2].

A sustained QAM waveform is generally modeled as a power signal rather than a finite-energy signal, so an ordinary finite ESD is not the appropriate description. For a sample path $x(t)$ observed through a duration- T rectangular window, its periodogram is [2]

$$P_{x,T}(f) = \frac{1}{T} |\hat{x}_T(f)|^2, \quad (9)$$

where $\hat{x}_T(f)$ is the Fourier transform of the windowed record. The periodogram estimates the distribution of average power over frequency, but its variance generally does not vanish as T increases; it is therefore not, by itself, a consistent PSD estimator. Low signal-to-noise ratio further obscures signal features [2]. Figure 7 illustrates the

periodogram of the 2-QAM signal from Figure 3.

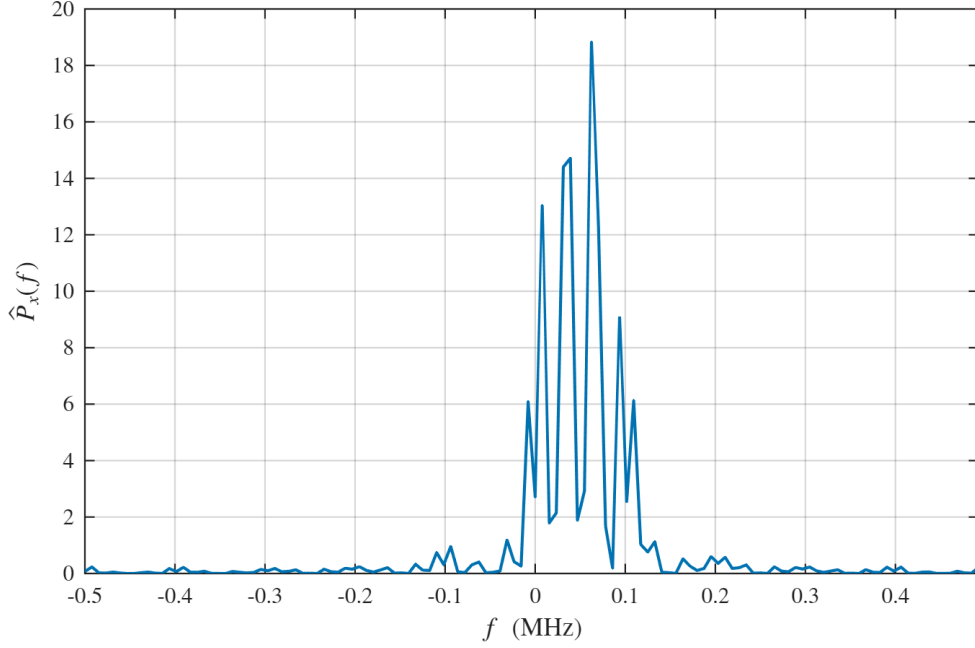


Figure 7: Squared-magnitude, power-scaled periodogram of the first 128 samples of the waveform $x(t)$ shown in Figure 3. The estimate uses a 128-point FFT.

The periodogram is roughly centered at f_c . Because this example uses rectangular pulses, its main lobe between the first spectral nulls has width approximately $2R_s$. Without prior knowledge, however, it would be challenging to infer the true CF and SR accurately from this one high-variance estimate.

One technique used to reduce this variability is the FSM [7]. Convolving the periodogram with a frequency-domain smoothing kernel $h(f)$ gives the estimate

$$\hat{S}_X(f) = h(f) * P_{x,T}(f), \quad (10)$$

where $*$ is the convolution operator. Frequency smoothing reduces estimator variance at the cost of spectral resolution and can introduce bias. In Figure 8, it makes the CF easier to estimate, but excessive smoothing broadens spectral features and can degrade SR estimation. The smoothing bandwidth must therefore be chosen according to the

bias–variance and resolution tradeoffs; no single minimum amount is optimal for every record.

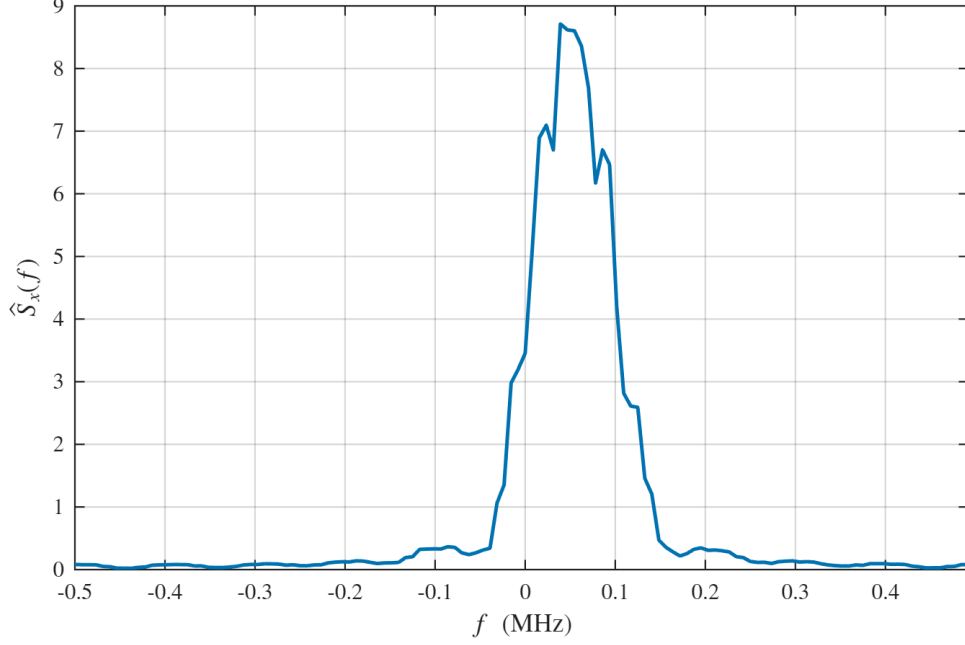


Figure 8: Frequency-smoothed PSD estimate of the waveform $x(t)$ shown in Figure 3. The 128-point periodogram is averaged over eight adjacent frequency bins.

2.2.2 The Cyclic Spectral Density Function

For a WSS random process, the theoretical PSD is the Fourier transform of $R_X(\tau)$. A finite-record periodogram estimates that PSD, and smoothing trades resolution for reduced variance. The analogous finite-record quantity for spectral correlation is the *cyclic periodogram*:

$$P_{x,T}^\alpha(f) = \frac{1}{T} \hat{x}_T\left(f + \frac{\alpha}{2}\right) \hat{x}_T^*\left(f - \frac{\alpha}{2}\right), \quad (11)$$

where $\hat{x}_T^*$ is the complex conjugate of $\hat{x}_T$. Its corresponding frequency-smoothed CSD estimate is

$$\hat{S}_X^\alpha(f) = h(f) * P_{x,T}^\alpha(f). \quad (12)$$

Across all values of cycle frequency α and spectral frequency f , the theoretical CSD $S_X^\alpha(f)$ characterizes the second-order periodic correlation structure of $X(t)$; it does not, in general, specify the process's higher-order statistics or full probability law [2]. Equation (12) is a finite-record estimate of that quantity. The CSD estimate of $X(t)$ is shown in Figure 9.

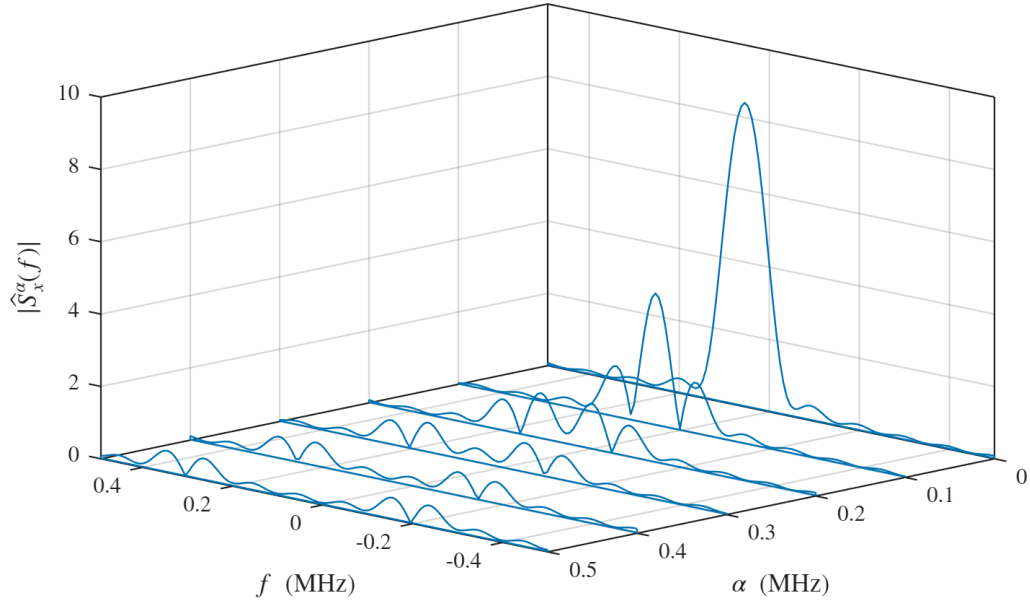


Figure 9: CSD magnitude estimate of the waveform $x(t)$ shown in Figure 3. A 128-point DFT is applied across the 31 estimated lags of $\hat{R}_x^\alpha(\tau)$, followed by averaging over eight adjacent frequency bins.

At $\alpha = 0$, the CSD reduces to the ordinary, zero-cycle power spectrum. Components at $\alpha \neq 0$ reveal periodic second-order structure; ideal stationary noise has no such nonzero-cycle components.

2.3 Spectral Correlation Analyzers

In this section, we move on to realizations of the CSD. The process of producing the CSD of a received waveform with no prior information is called *blind estimation*. Realizations of a blind estimator are often called *SCAs*. First, we discuss an example of a TSM-SCA as described by W. A. Gardner in the article *Cyclostationarity: Half a Century of Research* [3]. This TSM-SCA is used to blindly estimate the CSD of arbitrary (α, f) pairs.

We then consider the case in which the cycle frequency of interest equals the SR. M. V. Koch's technical report *Interference Mitigation Using Cyclic Autocorrelation and Multi-Objective Optimization* describes a *CAC-FFT algorithm* that forms $|x[n]|^2$, uses an FFT to evaluate the zero-lag CAC on a uniform rate grid, and selects candidate symbol rates from its peaks [5]. For each candidate rate, the report's implemented center-frequency detector systematically shifts and low-pass filters the signal segment, then evaluates a single-rate CAC at each offset. Koch also reports preliminary work on a different center-frequency preprocessor, which was not used in the reported system. The estimator in this thesis adapts the implemented CAC and carrier-sweep operations to a serial, fixed-filter hardware architecture; we refer to its pointwise statistic as the *CAF-DFT*.

2.3.1 Blind Cycle- and Spectral-Frequency Estimation

The most basic realization of the time-smoothed CSD is the TSM-SCA. A finite time-smoothed CSD estimate can be written as

$$\widehat{S}_X^\alpha(t, f)_{T, \Delta f} = \frac{1}{T} \int_{t-T/2}^{t+T/2} \Delta f X_{1/\Delta f}\left(u, f + \frac{\alpha}{2}\right) X_{1/\Delta f}^*\left(u, f - \frac{\alpha}{2}\right) du. \quad (13)$$

Here, $X_{1/\Delta f}(u, \nu)$ is the short-time Fourier transform centered at time u , with an effective duration of approximately $1/\Delta f$. The ideal CSD is approached by first letting $T \rightarrow \infty$ and then $\Delta f \rightarrow 0$; a finite estimate is statistically reliable only

when the time–bandwidth product $T\Delta f$ is sufficiently large. The TSM-SCA can be decomposed into multipliers, mixers, filters, and accumulators. To establish a baseline implementation, the block diagram based on Equation (13) is shown in Figure 10 [3].

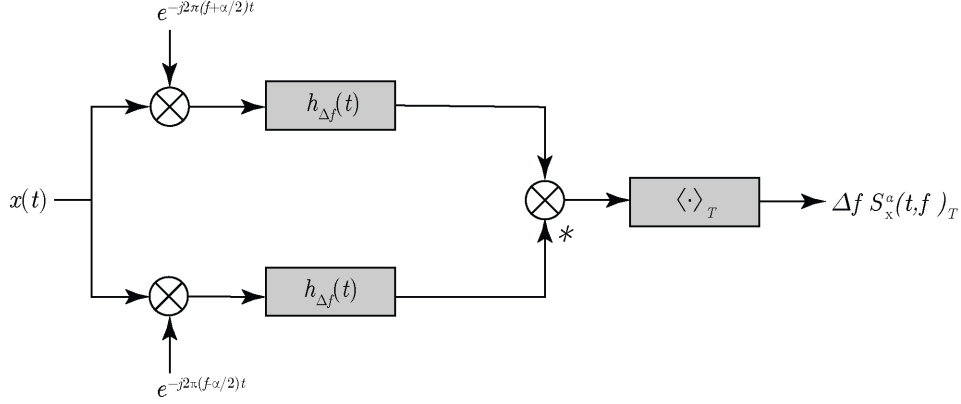


Figure 10: As described by Gardner in [3], this baseline SCA shifts the components centered at $f + \alpha/2$ and $f - \alpha/2$ to baseband. The component centers are separated by α . The shifted signals pass through low-pass filters $h_{\Delta f}(t)$ with bandwidth Δf and unity-passband height before their outputs are correlated, time-averaged over T , and scaled by Δf . The ideal CSD $S_X^\alpha(f)$ is obtained in the ordered limits $T \rightarrow \infty$ and then $\Delta f \rightarrow 0$.

Although the SCA shown in Figure 10 applies to an arbitrary waveform and gives the blind CSD estimate for any (α, f) pair, much of this search space is not helpful for most applications. If we can intelligently limit this search space, we can significantly improve the utility of the TSM-SCA.

2.3.2 Blind Symbol Rate-Center Frequency Point Estimation

Several common SCAs, such as the FSM, TSM, SSCA, and FAM, are discussed in [7]. Efficient digital implementations of several of these techniques use FFTs and block buffering, which can make them memory-intensive. Although larger memory allocations may be acceptable for terrestrial receivers, we must prioritize memory conservation for a CubeSat CR and explore alternative SCA designs.

Koch’s CAC-FFT uses peaks in the DFT of $|x[n]|^2$ to estimate candidate symbol

rates [5]. In the CAF-DFT used here, that operation is applied after channelization over candidate center frequencies. The relevant QAM-SRRC cycle-frequency structure follows from the linear-modulation model in Equation (4) [2]. Assume the data symbols are zero-mean and uncorrelated, with $E\{a_k a_m^*\} = \sigma_a^2 \delta_{km}$, and assume zero timing and carrier-phase offsets. The symmetric AF is then

$$R_X(t, \tau) = \sigma_a^2 e^{j2\pi f_c \tau} \sum_{k=-\infty}^{\infty} p\left(t + \frac{\tau}{2} - kT_0\right) p^*\left(t - \frac{\tau}{2} - kT_0\right). \quad (14)$$

Because Equation (14) is periodic with T_0 , we can express the CAF as follows:

$$R_X^\alpha(\tau) = \frac{\sigma_a^2}{T_0} e^{j2\pi f_c \tau} \int_{-\infty}^{\infty} p\left(u + \frac{\tau}{2}\right) p^*\left(u - \frac{\tau}{2}\right) e^{-j2\pi \alpha u} du, \quad \alpha = \frac{n}{T_0}, \quad n \in \mathbb{Z}. \quad (15)$$

The corresponding CSD is:

$$S_X^\alpha(f) = \frac{\sigma_a^2}{T_0} P\left(f - f_c + \frac{\alpha}{2}\right) P^*\left(f - f_c - \frac{\alpha}{2}\right), \quad \alpha = \frac{n}{T_0}. \quad (16)$$

Here, $P(f)$ is the Fourier transform of $p(t)$. If $p(t)$ is an SRRC pulse with positive roll-off, the nontrivial cycle frequencies for this CSD occur at $\alpha = \pm R_s$, in addition to the ordinary $\alpha = 0$ component. Since this implementation searches only positive cycle frequencies, it evaluates candidates near $S_X^{R_s}(f)$. In the ideal model, the peak coordinates estimate (R_s, f_c) to within the estimator's cycle- and spectral-frequency resolutions. This behavior is illustrated in Figure 11.

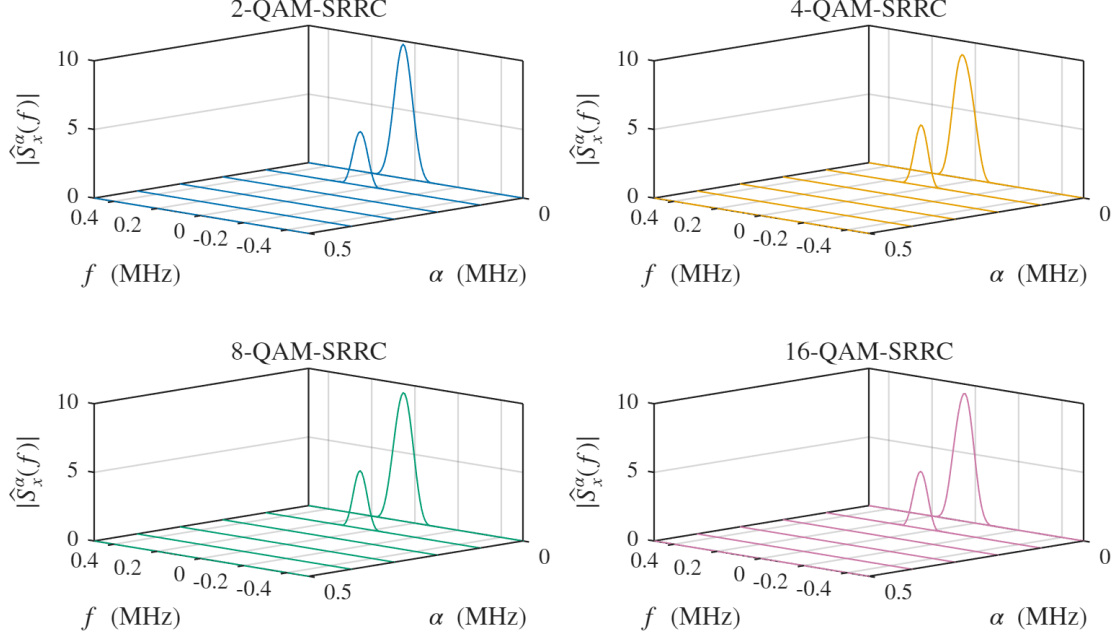


Figure 11: CSD magnitude estimates for independently generated 2-, 4-, 8-, and 16-QAM-SRRC waveforms with $R_s = 0.1$ MBd and $f_c = 0.05$ MHz. The $\alpha = 0$ slice is the ordinary PSD. For each modulation order, the nontrivial $\alpha = R_s$ slice has its maximum within one 7.8125-kHz DFT bin of f_c .

For the discrete CAF-DFT statistic used in this thesis, define the sequence in the k th channel as

$$z_k[n] = \left[\left(x[\cdot] e^{-j2\pi f_k(\cdot)/F_s} \right) * h[\cdot] \right] [n]. \quad (17)$$

Its estimated zero-lag CAF at the v th cycle-frequency candidate is

$$\hat{C}_x(\alpha_v, f_k) \triangleq \hat{R}_{z_k}^{\alpha_v}[0] = \frac{1}{N} \sum_{n=0}^{N-1} |z_k[n]|^2 e^{-j2\pi \alpha_v n / F_s}. \quad (18)$$

Here, α_v is the v th physical cycle-frequency candidate in hertz, f_k is the k th physical CF candidate, and n is the zero-based sample index. The quantity $\hat{C}_x(\alpha_v, f_k)$ is a channelized zero-lag CAF statistic—a filter-localized proxy for cyclic spectral content—rather than the point CSD value $\hat{S}_X^{\alpha_v}(f_k)$. On an N -point DFT-bin grid

with one-based v , $\alpha_v = (v - 1)F_s/N$, so the final kernel reduces to $e^{-j2\pi(v-1)n/N}$. More generally, an NCO can evaluate candidates that do not lie on that grid, so v is an array index rather than a frequency. A buffered, parallel realization of Equation (18) is shown in Figure 12; it adapts the mixer/filter search topology of the carrier-frequency sweep described in [5] while using a fixed $h[n]$ and no per-shift normalization.

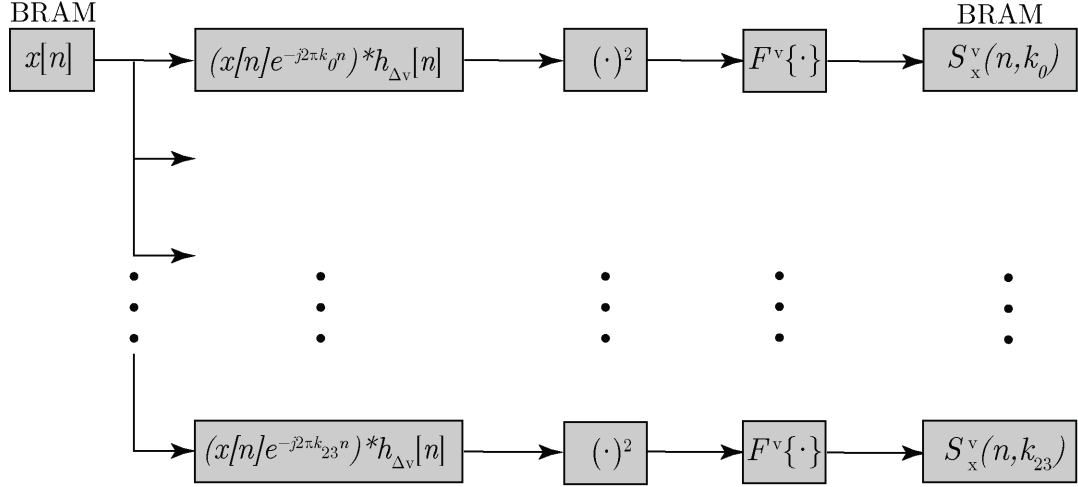


Figure 12: This buffered CAF-DFT realization frequency-shifts $x[n]$ by each CF candidate f_k and passes each shifted version through a low-pass filter, a process referred to as channelization. Evaluating the channelized autocorrelation at $\tau = 0$ gives the magnitude squared. Correlation with the cycle-frequency kernel at α_v , followed by division of the accumulated sum by N , produces the statistic $\hat{C}_x(\alpha_v, f_k)$. In the legacy diagram, k_i denotes the normalized frequency f_{k_i}/F_s , and $F^v\{\cdot\}$ denotes selection of the cycle-frequency coefficient.

As shown in Figure 12, the CAF-DFT narrows the search to candidate SR-CF pairs for M-QAM-SRRC waveforms. Fixing $\tau = 0$ removes the lag dimension and therefore avoids the Fourier transform across lag required to form a full CSD. It does not eliminate the cycle-frequency correlation in Equation (18), which may be implemented by a DFT or by selected NCO kernels. This structure lends itself well to streaming processing.

2.4 Spectrum Sensing

Spectrum sensing is the process of periodically monitoring a specific frequency band to infer the presence or absence of users. Monitoring is typically performed by a secondary user (SU) to reuse inactive frequencies allocated or licensed to a primary user (PU) [4, 6]. These allocated, inactive areas of a band are referred to as *spectral holes* [4]. In this section, we discuss several detection techniques and the strengths of CSP for spectrum sensing.

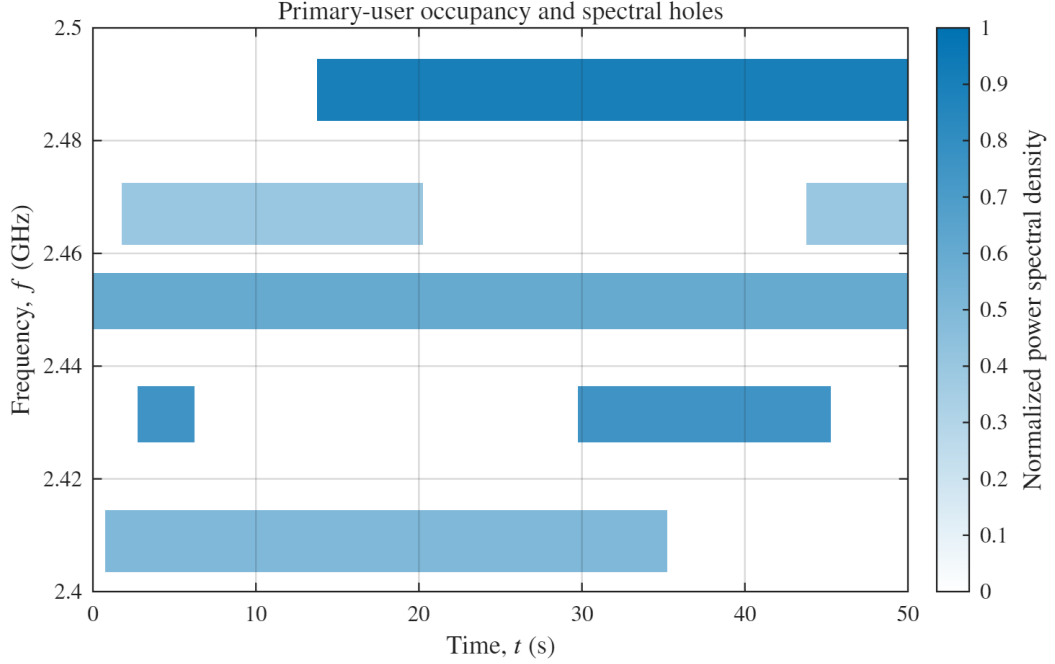


Figure 13: Conceptual time–frequency occupancy of five PU bands. White regions are spectral holes and blue regions indicate normalized PU power. An SU that maintains awareness of active PUs can hop among these holes and adapt its symbol rate to the currently available bandwidth.

2.4.1 The Binary Hypothesis Test

Let us consider an SU that is monitoring a narrow RF band for the presence of a PU. There are two hypotheses: under H_0 , the PU is inactive; under H_1 , the PU is active [6]. The received signal is written as follows:

$$H_0 : Y(t) = \eta(t) \quad (19)$$

$$H_1 : Y(t) = G(t) * X(t) + \eta(t), \quad (20)$$

where $X(t)$ denotes the transmitted signal from the primary user, $G(t)$ is the channel impulse response, $*$ denotes convolution, and $\eta(t)$ is zero-mean additive white Gaussian noise (AWGN) independent of $X(t)$. In the sampled model used later, $\eta[n]$ is represented by IID Gaussian samples. Exactly one of H_0 and H_1 is true, and the receiver selects a decision $\hat{H}$ from the observed samples [6]. The four conditional decision probabilities are

1. Probability of correct rejection, $P_{\text{CR}} = \Pr(\hat{H} = H_0 \mid H_0)$: H_0 is true; choose H_0 .
2. Probability of miss, $P_{\text{M}} = \Pr(\hat{H} = H_0 \mid H_1)$: H_1 is true; choose H_0 .
3. Probability of false alarm, $P_{\text{FA}} = \Pr(\hat{H} = H_1 \mid H_0)$: H_0 is true; choose H_1 .
4. Probability of detection, $P_{\text{D}} = \Pr(\hat{H} = H_1 \mid H_1)$: H_1 is true; choose H_1 .

Outcomes P_{D} and P_{CR} correspond to correct decisions, whereas P_{M} and P_{FA} correspond to errors. If the SU transmits only after deciding H_0 , these probabilities have the following physical interpretations:

1. P_{CR} is the probability that the SU correctly identifies an idle band and may transmit.
2. P_{M} is the probability that the SU declares an occupied band idle and may interfere with the PU.
3. P_{FA} is the probability that the SU declares an idle band occupied and forgoes an opportunity to transmit.

4. P_D is the probability that the SU correctly identifies an active PU and refrains from transmitting.

For simple hypotheses with known conditional densities, the Neyman–Pearson lemma states that a likelihood-ratio test maximizes P_D among tests whose P_{FA} does not exceed a specified level [6, 8]. Let $\mathbf{y}$ denote the observed sample vector. Its likelihood ratio is

$$\Lambda(\mathbf{y}) = \frac{p(\mathbf{y} \mid H_1)}{p(\mathbf{y} \mid H_0)}. \quad (21)$$

Comparing $\Lambda(\mathbf{y})$ with a threshold γ gives the likelihood-ratio test

$$\Lambda(\mathbf{y}) \underset{H_0}{\overset{H_1}{\gtrless}} \gamma. \quad (22)$$

If $\Lambda(\mathbf{y})$ exceeds γ , choose H_1 ; otherwise, choose H_0 [6]. Figure 14 illustrates this concept:

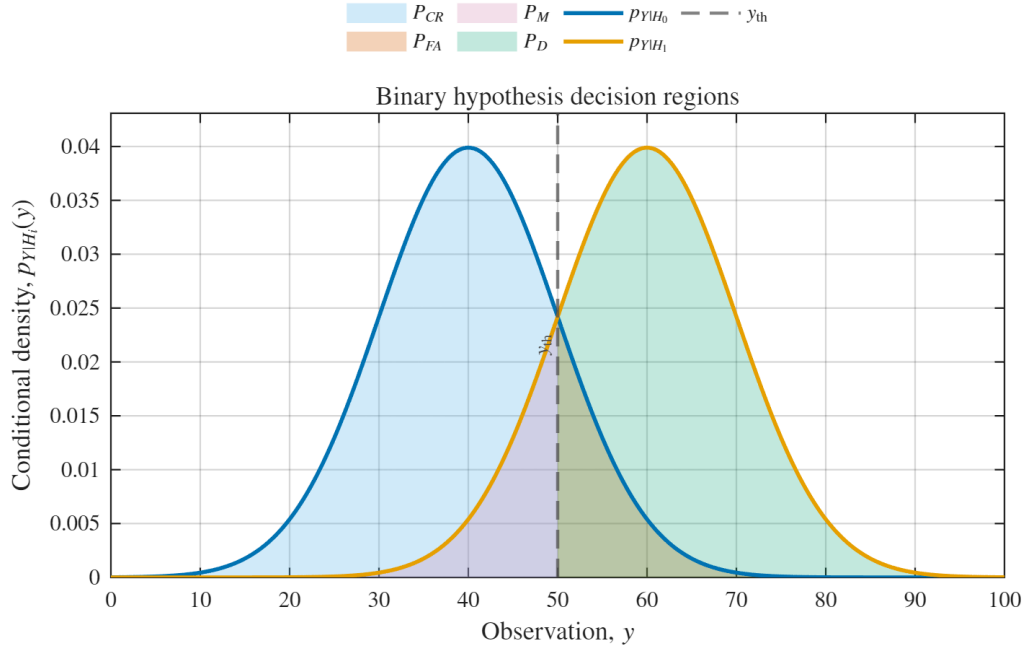


Figure 14: Conditional Gaussian observation densities with equal priors, equal decision costs, common standard deviation 10, and means 40 and 60. The observation threshold $y_{\text{th}} = 50$ corresponds to likelihood-ratio threshold $\gamma = 1$ and gives $P_{\text{FA}} = 0.1587$ and $P_{\text{D}} = 0.8413$. Shading identifies the four decision regions.

2.4.2 Classical Detection Methods

There are many spectrum-sensing detection methods available. They are broadly categorized as *cooperative*—where coordination among many receivers is leveraged to detect a waveform—or *non-cooperative*—where the receiver detects based only on what it can sense [4]. In this subsection, we discuss three non-cooperative detection methods.

2.4.2.1 Matched Filter Detector

For a deterministic signal known to the receiver and observed in AWGN, the matched-filter test is optimal [4, 6]. The matched filter maximizes the output signal-to-noise ratio (SNR) by correlating the received signal with the known waveform and is often used within a coherent detector. Its performance degrades when the received

waveform differs from the template or contains unmodeled interference [4].

2.4.2.2 Energy Detector

For a single-antenna sampled model in which the signal and noise samples are IID zero-mean Gaussian random variables and the noise power is known, energy detection is optimal [6]. The energy detector compares the accumulated received energy with a threshold based on the noise power, declaring a PU present when the threshold is exceeded [4, 6]. It is attractive when little prior information about the signal is available, but it does not classify signal features and is vulnerable to unknown or changing noise levels.

2.4.2.3 Cyclostationary Feature Detector

A cyclostationary feature detector (CFD) can distinguish a modulated signal from stationary noise by testing for nonzero-cycle features, often at lower SNR than an energy detector can support [4]. Let $Q_Y(\alpha_v, f_k)$ denote either a full-CSD value $S_Y^{\alpha_v}(f_k)$ or the channelized zero-lag CAF quantity $C_Y(\alpha_v, f_k)$ underlying Equation (18). For either feature, stationary AWGN gives the idealized population relations

$$H_0 : Q_Y(\alpha_v, f_k) = 0, \quad \alpha_v \neq 0, \quad (23)$$

$$H_1 : Q_Y(\alpha_v, f_k) \neq 0 \quad \text{for some } (\alpha_v, f_k). \quad (24)$$

These equalities describe the ideal nonzero-cycle feature. A finite-sample estimate $\hat{Q}_y(\alpha_v, f_k)$ is generally not exactly zero under H_0 because of estimation variance and spectral leakage, so the decision requires a threshold. If the statistic exceeds that threshold at any f_k for a fixed α_v , the receiver declares cyclostationary behavior at that candidate rate. Under the assumed M-QAM-SRRC, single-feature model, a uniquely resolved maximizing f_k estimates the PU's center frequency.

2.4.3 Test Statistic and Detection Threshold

In a spectrum-sensing environment, the receiver may lack prior information about the signal and its activity probability [6]. For the illustrative detector used here, define a nonnegative pointwise test statistic $\ell(\hat{Q}_y(\alpha_v, f_k))$, or simply ℓ , for each candidate (α_v, f_k) pair:

$$\ell = |\hat{Q}_y(\alpha_v, f_k)|^2. \quad (25)$$

The statistic still requires a threshold γ . Noise power is difficult to estimate in a realistic spectrum-sensing environment [6], so this demonstration instead normalizes the threshold to the estimated average received power,

$$\hat{P}_y = \frac{1}{N} \sum_{n=0}^{N-1} |y[n]|^2. \quad (26)$$

The empirical threshold is $\gamma = \beta \hat{P}_y^2$, where the dimensionless scale β controls sensitivity. It is not itself a calibrated false-alarm probability; such calibration would require the statistic's null distribution or Monte Carlo trials. The resulting test is

$$|\hat{Q}_y(\alpha_v, f_k)|^2 \underset{H_0}{\overset{H_1}{\geq}} \beta \left(\frac{1}{N} \sum_{n=0}^{N-1} |y[n]|^2 \right)^2. \quad (27)$$

Any nontrivial cycle-frequency slice containing a value above the threshold in Equation (27) is considered detected. Under the assumed signal model and a uniquely resolved peak, the largest value within each detected slice estimates its center frequency. Figure 15 illustrates this rule using a full-CSD estimate, $\hat{Q}_y = \hat{S}_Y$; Chapter III applies it to the channelized statistic $\hat{Q}_y = \hat{C}_y$.

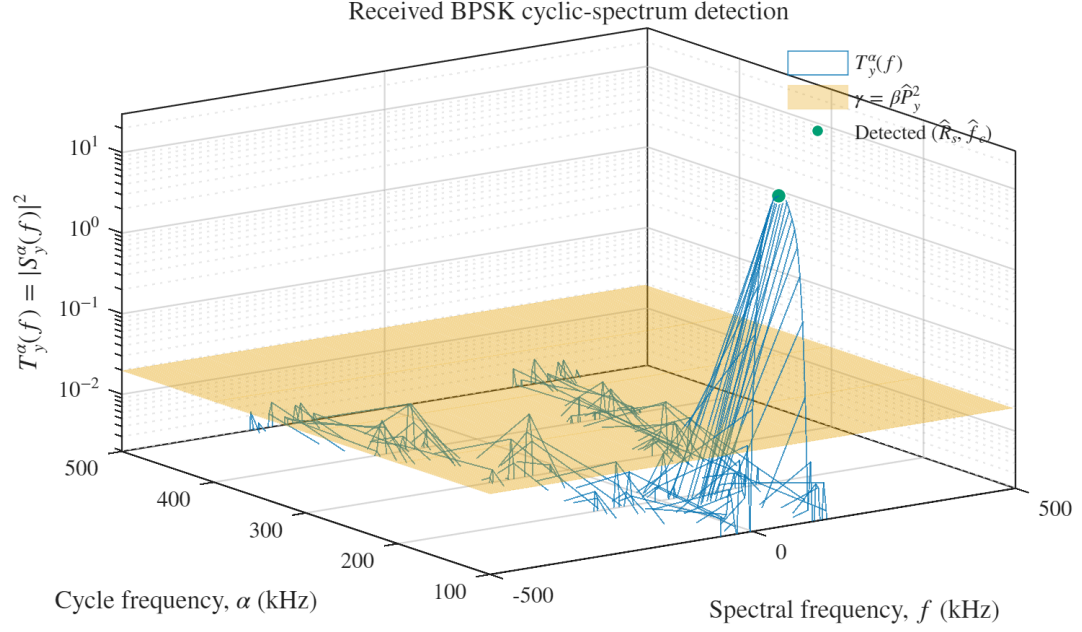


Figure 15: Full-CSD test statistic for a received BPSK-SRRC waveform with $R_s = 100$ kBd, $f_c = 50$ kHz, and requested $E_b/N_0 = 14$ dB. The nontrivial cycle-frequency slices are compared with the flat empirical threshold $\gamma = 10^{-2} \hat{P}_y^2$. Only the 100-kHz slice exceeds the threshold, and its maximum occurs at 50 kHz.

CHAPTER III

IMPLEMENTATION

This work studies spectrum sensing for a prospective CubeSat cognitive radio (CR). Within the deliberately limited M-QAM-SRRC model used in this thesis, the receiver searches a specified grid of candidate symbol rates (SRs) and center frequencies (CFs) and uses cyclostationary feature detection (CFD) to distinguish modeled signals from noise. A spectral correlation analyzer (SCA) can realize this detector. Conventional FFT-based SCAs obtain high throughput by evaluating many spectral channels in parallel. Their implementations buffer blocks of complex samples and intermediate transform results, so storage grows with both the capture length and the number of retained channels. This memory can compete with limited on-chip FPGA block RAM (BRAM) in a CubeSat-class implementation.

BRAM capacity is therefore a central design constraint. Reducing buffered storage improves fit within on-chip memory, while sample-by-sample time-domain processing trades the parallel throughput of an FFT-based analyzer for a deeply pipelined data path. To limit the resulting search time, the application is restricted to detecting the SR and CF of M-QAM-SRRC waveforms rather than estimating every cyclic feature of an arbitrary waveform. For this blind SR–CF estimator, we borrow the communications term *streaming*. Thus, this design will be referred to as the *streaming SCA*. It is an FPGA architecture study, not a flight-qualified implementation.

3.1 Motivation

Orbital systems may use a single spacecraft, a constellation, or a distributed architecture such as a fractionated, clustered, or trailing formation. Distributed architectures coordinate multiple spacecraft, and CubeSats are one relevant small-spacecraft platform [1].

Crosslinks and relays create operating scenarios in which inter-satellite interference may need to be monitored. Spectrum sensing is one candidate mechanism for detecting occupied channels and informing an interference-avoidance response.

As discussed in the previous chapter, SCAs estimate cyclic spectral structure that can be used for spectrum sensing [7]. Several efficient digital SCA realizations use FFTs, sample blocks, and intermediate arrays whose storage requirements can be substantial [6, 7].

For a concrete scaling example, an SSCA intermediate array with $N = 2^{16}$ time samples and $N_p = 32$ retained frequency channels contains NN_p complex words [7]. At 16 bits for each in-phase and quadrature component, or 32 bits per complex word, that array alone requires $2^{16} \times 32 \times 32 = 67,108,864$ bits, or 67.1 Mbit, before control and buffering overhead. The recovered 2021 component design reports and project-recreation scripts consistently identify the intended development platform as a ZedBoard with an XC7Z020CLG484-1. That device provides 4.9 Mbit of block RAM [9], so this particular array would not fit wholly in its BRAM without reducing the stored data or changing the architecture. This establishes the documented component-development target; it does not establish the provenance of the experiment-level resource counts in Chapter IV, whose top-level synthesis report is unavailable.

3.2 The Streaming SCA

Let us assume that we wish to sense modeled waveforms present in a channel. As discussed in Chapter II, the single-term CAF-DFT form can evaluate selected

frequency candidates without a block FFT. To create a low-memory design suitable for evaluation on the selected FPGA, we alter the CAF-DFT such that it:

- Evaluates only one pipelined (v, k) pair at a time, called a point. This is equivalent to one branch in Figure 12. If our SCA is well designed, the branches can be reconfigured so that multiple instances can be used in parallel when the platform has extra resources.
- Avoids a full-capture BRAM buffer. A block-based realization buffers a capture so that every (v, k) pair is evaluated on the same samples. This design instead accepts a continuous input sample stream without retaining the full capture; ADC interfacing is outside the module. Consequently, different candidate pairs can observe different time intervals, but the design reduces buffered storage when the signal is sufficiently stable over the search.
- Targets FPGA implementation so that the estimator can use deterministic pipelines and configurable parallelism within the platform’s resource budget. FPGA implementation alone does not establish radiation tolerance; device qualification and mitigation are separate design considerations.
- Does not evaluate the zero-cycle-frequency case $\alpha = 0$, which corresponds to the ordinary PSD rather than a nontrivial cyclostationary feature.
- Uses a fixed two-tap low-pass filter (LPF), corresponding to taps $[0.5, 0.5]$. Koch’s evaluated carrier-frequency detector systematically shifts each segment, dynamically configures the LPF bandwidth to the target symbol rate, and evaluates the CAC at that rate [5]. The streaming design is a serial hardware adaptation of that sweep; to avoid a reconfigurable coefficient bank, it uses the same two coefficients for every candidate.

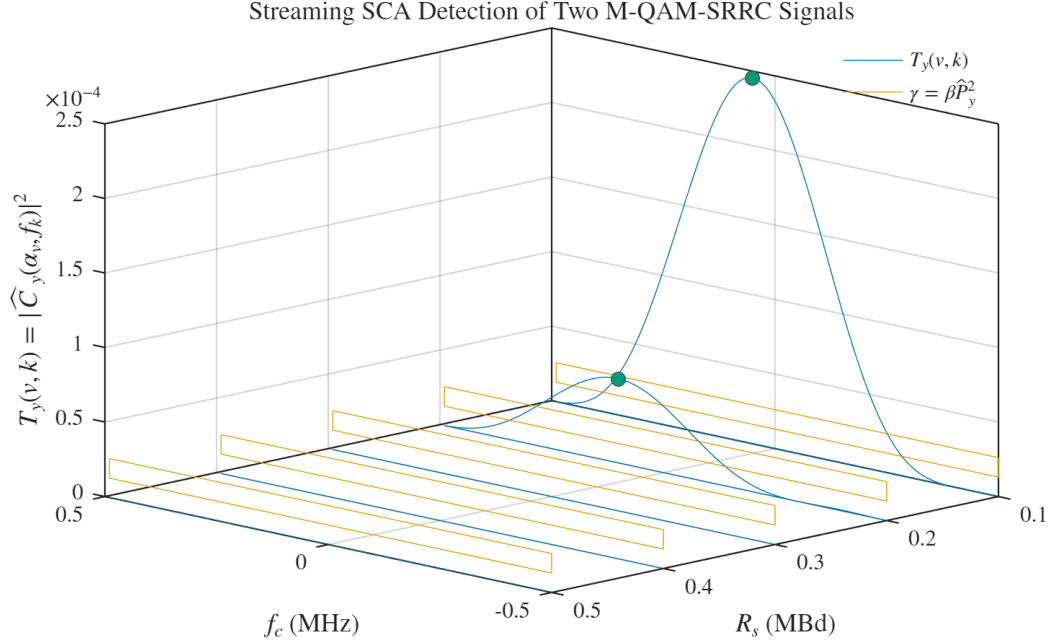


Figure 16: Streaming SCA test statistic $T_y(v, k) = |\hat{C}_y(\alpha_v, f_k)|^2$ for two M-QAM-SRRC signals in a shared noise floor. The transmitted parameter pairs are $(R_s, f_c) = (100 \text{ kBd}, 50 \text{ kHz})$ and $(200 \text{ kBd}, 100 \text{ kHz})$. Both pairs lie on the estimator grid, both row maxima recover the transmitted CF, and both exceed the constant threshold $\gamma = \beta \hat{P}_y^2$ with $\beta = 10^{-3}$. Green markers indicate the detected row maxima.

The streaming SCA was modeled in MATLAB to produce Figure 16. The two generated components are scaled so that a single receiver-noise realization gives the requested E_b/N_0 values of 14 dB and 9 dB. A common automatic-gain-control (AGC) scale factor then limits the composite waveform without changing either component's carrier-to-noise ratio. On the 10 kHz CF grid, the row maxima occur at the transmitted CFs of 50 kHz and 100 kHz; their test-statistic-to-threshold ratios are 9.41 and 2.16, respectively. These numerical values are reproducible because the model fixes its random-number seed. The executable MATLAB model is imported directly in Appendix A.

In exchange for this low memory use, the processing time is increased. With one (v, k) point evaluated at a time, no backpressure, and a sustained rate of one

valid sample per clock, an exhaustive search over N_v SR candidates and N_k CF candidates requires approximately NN_vN_k sample clocks, in addition to pipeline and control overhead. For fixed search spans, the processing delay therefore increases with capture length and with the number of candidates:

- SR step size: A smaller step produces more SR candidates and a longer search, whereas a larger step coarsens the grid and can increase parameter mismatch.
- CF step size: A smaller step produces more CF candidates and a longer search, whereas a larger step coarsens the grid and can increase parameter mismatch.
- Capture length N : A larger capture takes longer to process but provides a longer observation interval; a smaller capture reduces delay at the cost of a less stable finite-record estimate.

3.3 Digital Design

To describe the implementation of the streaming SCA as a digital circuit, we will walk through each component and its subcomponents. The documented signal-processing datapath uses custom VHDL modules rather than a vendor FFT core. MATLAB and Xilinx Vivado were nevertheless used for modeling and implementation, and inferred device resources can affect portability. The tables summarize the configured documentation interface rather than reproducing every literal entity generic or port name; in particular, some propagation-only width and latency generics are omitted, and several arithmetic names are shortened. The port tables retain the RTL’s ready/valid names, but the recovered modules form fixed-latency pipelines rather than a fully elastic ready/valid interface: several data paths and state machines do not stall when `SampleOutReady` is low. The characterization therefore assumes continuous, non-backpressured operation with one valid sample per clock and the downstream path held ready. Supporting arbitrary bubbles or backpressure requires revised control logic and re-verification.

The recovered archive does not contain the exact thesis-era source revision. Its Git index identifies the later revision `1972ccba` from July 5, 2022, but the extracted worktree contains tracked divergences and its pre-clone Git object pack is missing, preventing reconstruction of the 2021 history. A contemporaneous CFE design report explicitly documents a two-tap configuration in which `TAP_COUNT=2` produces the coefficients $[0.5, 0.5]$. By contrast, the active later source uses a 41-tap, 128-bank filter. The archive therefore substantiates the thesis configuration’s design intent but cannot establish bit-true conformance of the historical VHDL. It also contains a later, incomplete top-level wrapper with an FFT-based symbol-rate preprocessor; that unfinished development branch is outside the implementation characterized here. The discussion occasionally refers to VHDL functions, such as `resize()`; however, the signal-processing description is otherwise language-agnostic.

3.3.1 Center Frequency Estimator

The purpose of the center frequency estimator (CFE) is to produce a test-statistic surface over the SR–CF candidates. Let v and k index user-defined candidate arrays with physical frequencies α_v and f_k , respectively. For an input sequence $x[n]$, define

$$z_k[n] = \left[\left(x[\cdot] e^{-j2\pi f_k(\cdot)/F_s} \right) * h[\cdot] \right] [n], \quad \hat{C}_x(\alpha_v, f_k) = \frac{1}{N} \sum_{n=0}^{N-1} |z_k[n]|^2 e^{-j2\pi \alpha_v n / F_s}.$$

The center-frequency control word configures a downconverting local oscillator for the factor $e^{-j2\pi f_k n / F_s}$. Next, in the thesis configuration, an LPF module smooths the mixed signal with the fixed taps $[0.5, 0.5]$. The SRE then evaluates the zero-lag CAF coefficient $\hat{C}_x(\alpha_v, f_k) = \hat{R}_{z_k}^{\alpha_v}[0]$, and the final magnitude-squared operation produces one point of the test statistic, $T_x(v, k) = |\hat{C}_x(\alpha_v, f_k)|^2$. Thus, v and k are candidate-array indices rather than frequencies themselves. As explained in Chapter II, $\hat{C}_x$ is a filter-localized cyclic statistic, not literally the CSD value at f_k .

For a W -bit phase accumulator, a requested frequency f is represented by the integer frequency-control word $\text{round}(2^W f/F_s)$, modulo 2^W , so the frequency quantum is $F_s/2^W$ and nearest-word error is at most half that quantum. In the thesis configuration, $W = 16$ and $F_s = 1$ MHz, giving a quantum of approximately 15.26 Hz. The center-frequency port is signed to represent the range $-F_s/2$ through $F_s/2 - F_s/2^W$. The symbol-rate port is declared unsigned, but the surviving SRE passes its bit pattern to a signed NCO input; therefore only codes below 2^{W-1} retain a positive signed interpretation, while the 2^{W-1} code represents the equivalent Nyquist endpoint $-F_s/2$. The searches in this thesis request rates no greater than $F_s/2$. Although the RTL exposes width and lookup-phase generics, literal mixer bit slices mean the surviving implementation only evidences the $W = 16$ configuration, and the lookup-phase generic is unused. If the cycle-frequency candidates are restricted to an N -point DFT grid, MATLAB's one-based index satisfies $\alpha_v = (v - 1)F_s/N$; the hardware frequency-control word also permits candidates that are not on that grid, subject to its quantization.

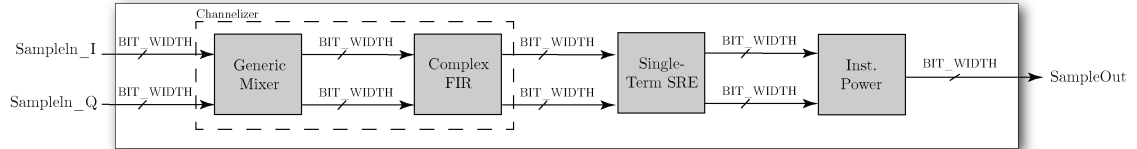


Figure 17: Digital design of the CFE module.

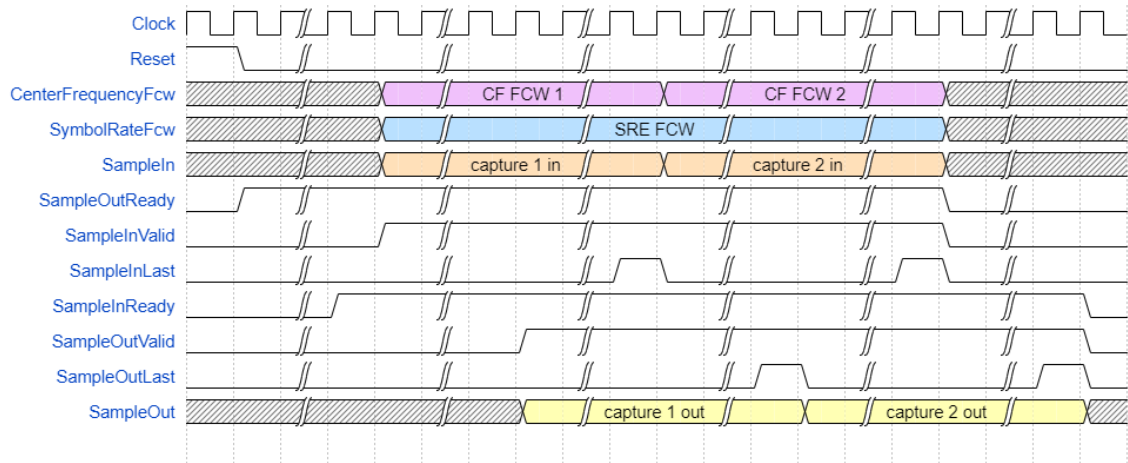


Figure 18: Timing diagram of the CFE module.

Table I: Configured CFE interface.

Port Name	Direction	Data Type	Description
N_SAMPLES	Generic	natural	Requested length; ignored by the accumulator.
BIT_WIDTH	Generic	natural	Sample width: one sign bit and BIT_WIDTH-1 fractional bits.
Clock	Input	std_logic	Synchronous clock.
Reset	Input	std_logic	Synchronous, active-high reset.
SampleOutReady	Input	std_logic	Nominal downstream-ready input; does not stall.
SampleInValid	Input	std_logic	High when input is valid.
SampleInLast	Input	std_logic	Marks boundaries between CF-SR pairs being processed.
CenterFrequencyFcw	Input	signed(15 downto 0)	Signed 16-bit CF control word.
SymbolRateFcw	Input	unsigned(15 downto 0)	Unsigned 16-bit SR control word.
SampleIn_I	Input	signed(BIT_WIDTH-1 downto 0)	Real component of sample.
SampleIn_Q	Input	signed(BIT_WIDTH-1 downto 0)	Imaginary component of sample.
SampleInReady	Output	std_logic	Nominal upstream-ready output; no acceptance handshake.
SampleOutValid	Output	std_logic	High when output is valid.
SampleOutLast	Output	std_logic	Marks boundaries between CF-SR pairs being processed.
OverflowStatus	Output	std_logic_vector(3 downto 0)	[3]: CF mixer I [2]: CF mixer Q [1]: SR mixer I [0]: SR mixer Q
SampleOut	Output	unsigned(2*BIT_WIDTH-1 downto 0)	Nonnegative magnitude-squared result.

3.3.2 Symbol Rate Estimator

The purpose of the SR estimator (SRE) is to estimate the correlation between a waveform and the candidate SR cycle frequency α_v . Up to the fixed-point scaling and archived accumulator limitations described in Section 3.3.9, the target computation is the finite-record CAF estimate evaluated at $\tau = 0$:

$$\hat{R}_X^{\alpha_v}[0] = \frac{1}{N} \sum_{n=0}^{N-1} |X[n]|^2 e^{-j2\pi\alpha_v n/F_s}.$$

The instantaneous-power module is intended to compute $|X[n]|^2$. The single-frequency Fourier-sum module then evaluates the displayed zero-lag CAF expression at α_v . The surviving SRE snapshot, however, configures the power block to retain unsigned bits 29 through 14 and then reinterprets that 16-bit word as signed at the DFT input. For 16-bit inputs with 15 fractional bits, powers at or above approximately 0.5 can therefore acquire a negative signed interpretation, and an ideal power of 1.0 is carried by a discarded higher bit and maps to zero. The unsigned-power-to-signed interface and truncation must be corrected and re-verified across the intended input range before the RTL can be considered bit-true to the displayed equation.

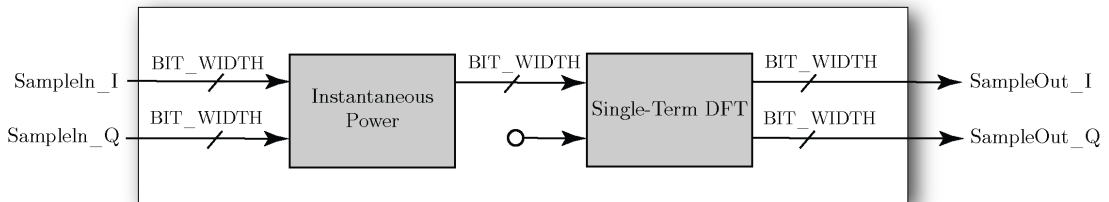


Figure 19: Digital design of the SRE module.

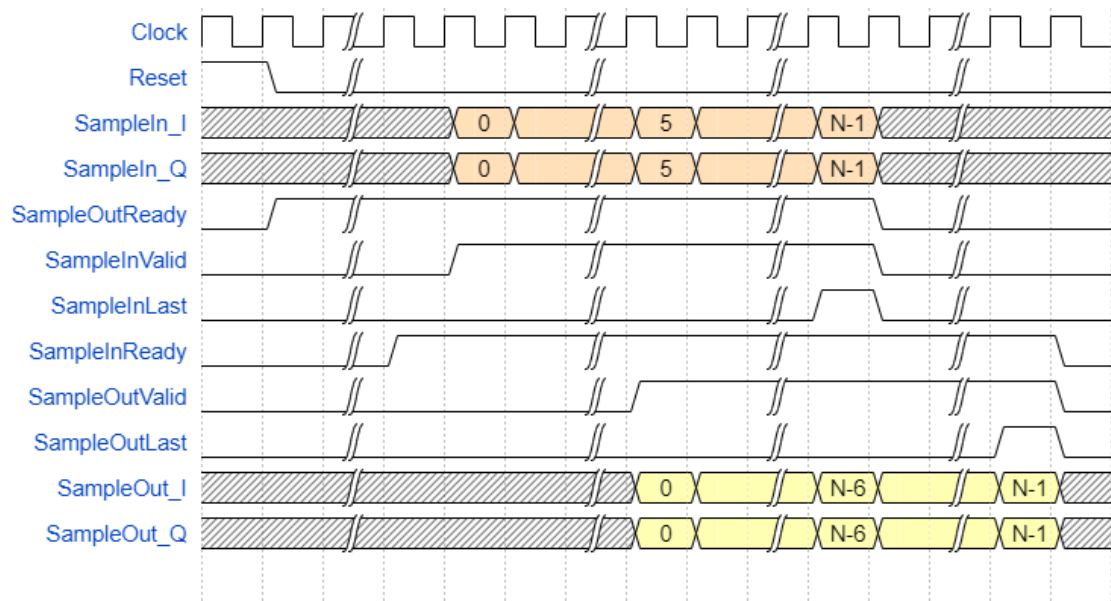


Figure 20: Timing diagram of the SRE module.

Table II: Configured SRE interface.

Port Name	Direction	Data Type	Description
N_SAMPLES	Generic	natural	Requested length; ignored by the accumulator.
BIT_WIDTH	Generic	natural	Sample width: one sign bit and BIT_WIDTH-1 fractional bits.
Clock	Input	std_logic	Synchronous clock.
Reset	Input	std_logic	Synchronous, active-high reset.
SampleOutReady	Input	std_logic	Nominal downstream-ready input; does not stall.
SampleInValid	Input	std_logic	High when input is valid.
SampleInLast	Input	std_logic	Marks boundaries between SRs being processed.
OscillatorFcw	Input	unsigned(15 downto 0)	Unsigned 16-bit SR control word.
SampleIn_I	Input	signed(BIT_WIDTH-1 downto 0)	Real component of sample.
SampleIn_Q	Input	signed(BIT_WIDTH-1 downto 0)	Imaginary component of sample.
SampleInReady	Output	std_logic	Nominal upstream-ready output; no acceptance handshake.
SampleOutValid	Output	std_logic	High when output is valid.
SampleOutLast	Output	std_logic	Marks boundaries between SRs being processed.
OverflowStatus	Output	std_logic_vector(1 downto 0)	[1]: mixer I [0]: mixer Q
SampleOut_I	Output	signed(BIT_WIDTH-1 downto 0)	Real component of result.
SampleOut_Q	Output	signed(BIT_WIDTH-1 downto 0)	Imaginary component of result.

3.3.3 Discrete Fourier Transform

The DFT is used to perform frequency-domain analysis of discrete-time signals. This implementation evaluates a finite-record Fourier sum at one selected physical frequency f_0 at a time:

$$\hat{X}(f_0) = \sum_{n=0}^{N-1} X[n]e^{-j2\pi f_0 n/F_s}.$$

The standard N -point DFT coefficient $\hat{X}[\ell]$ is the special case $f_0 = \ell F_s/N$ for the zero-based bin $0 \leq \ell < N$. In the intended hardware operation, the oscillator frequency-control word selects f_0 , the mixer forms the complex products $X[n]e^{-j2\pi f_0 n/F_s}$, and the accumulator sums N products. The quarter-wave lookup-table reconstruction can introduce a constant NCO initial-phase factor, but that factor cancels in the final magnitude squared. For the SRE, $f_0 = \alpha_v$; when the candidates lie on the DFT grid, $\ell = v - 1$ under MATLAB's one-based indexing. Section 3.3.9 documents a boundary discrepancy in the surviving accumulator RTL.

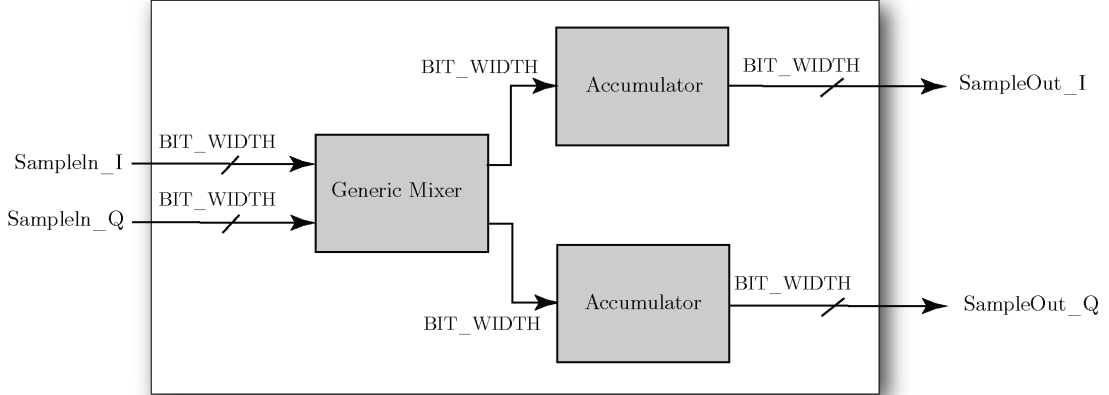


Figure 21: Block diagram of the discrete Fourier transform module.

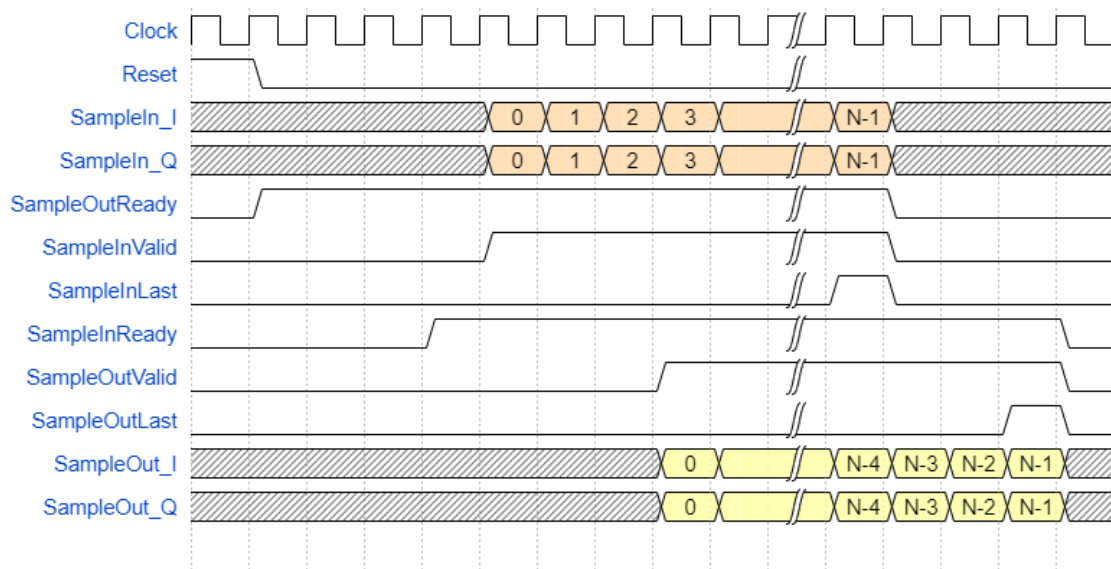


Figure 22: Timing diagram of the DFT module.

Table III: Configured DFT interface.

Port Name	Direction	Data Type	Description
N_SAMPLES	Generic	natural	Requested length; ignored by the accumulator.
BIT_WIDTH	Generic	natural	Sample width: one sign bit and BIT_WIDTH-1 fractional bits.
Clock	Input	std_logic	Synchronous clock.
Reset	Input	std_logic	Synchronous, active-high reset.
SampleOutReady	Input	std_logic	Nominal downstream-ready input; does not stall.
SampleInValid	Input	std_logic	High when input is valid.
SampleInLast	Input	std_logic	Marks boundaries between frequencies being processed.
OscillatorFcw	Input	signed(15 downto 0)	Signed 16-bit frequency control word.
SampleIn_I	Input	signed(BIT_WIDTH-1 downto 0)	Real component of sample.
SampleIn_Q	Input	signed(BIT_WIDTH-1 downto 0)	Imaginary component of sample.
SampleInReady	Output	std_logic	Nominal upstream-ready output; no acceptance handshake.
SampleOutValid	Output	std_logic	High when output is valid.
SampleOutLast	Output	std_logic	Marks boundaries between frequencies being processed.
OverflowStatus	Output	std_logic_vector(1 downto 0)	[1]: mixer I [0]: mixer Q
SampleOut_I	Output	signed(BIT_WIDTH-1 downto 0)	Real component of result.
SampleOut_Q	Output	signed(BIT_WIDTH-1 downto 0)	Imaginary component of result.

3.3.4 Frequency Mixer

Frequency translation is performed by multiplying the input by a complex local oscillator generated by a numerically controlled oscillator (NCO). For a positive-frequency component at f_1 , multiplication by the downconverting oscillator at f_{LO} gives

$$e^{j2\pi f_1 n/F_s} e^{-j2\pi f_{LO} n/F_s} = e^{j2\pi(f_1 - f_{LO})n/F_s}.$$

When $f_{LO} = f_1$, the component is translated to zero frequency. The implementation generates the oscillator with a phase accumulator and a sine/cosine lookup table; the frequency-control word sets its phase increment. In the archived mixer RTL, the phase accumulator advances on every clock rather than only on accepted `SampleInValid` transactions. The implemented model therefore assumes one accepted sample per clock. If a low-valid clock represents a genuinely elapsed but unobserved sample interval, the phase advance is physically consistent but the accepted samples are no longer consecutively indexed as in the equations; if it represents a stalled sample, the phase update must instead be gated with the sample-accept event.

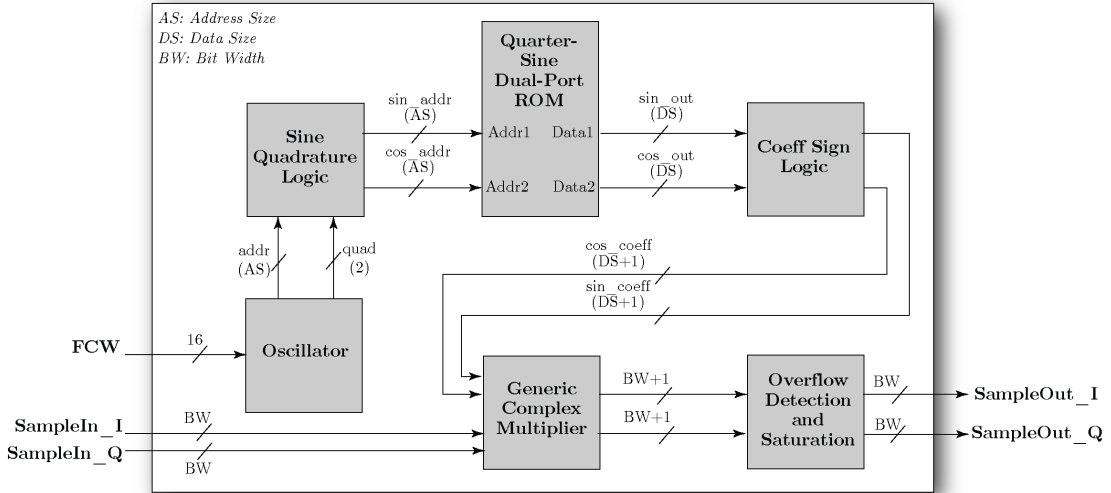


Figure 23: Digital design of the frequency mixer module.

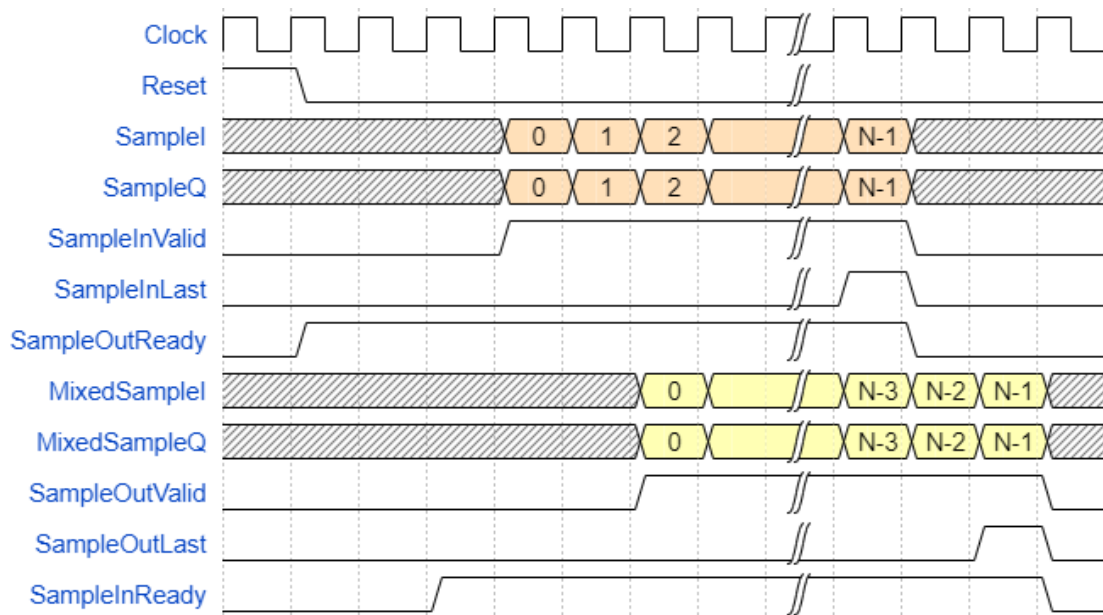


Figure 24: Timing diagram of the frequency mixer module.

Table IV: Configured frequency-mixer interface.

Port Name	Direction	Data Type	Description
BIT_WIDTH	Generic	natural	Sample width: one sign bit and BIT_WIDTH-1 fractional bits.
Clock	Input	std_logic	Synchronous clock.
Reset	Input	std_logic	Synchronous, active-high reset.
SampleOutReady	Input	std_logic	Nominal downstream-ready input; does not stall.
SampleInValid	Input	std_logic	High when input is valid.
SampleInLast	Input	std_logic	Pass through.
OscillatorFw	Input	signed(15 downto 0)	Signed 16-bit frequency control word.
SampleIn_I	Input	signed(BIT_WIDTH-1 downto 0)	Real component of sample.
SampleIn_Q	Input	signed(BIT_WIDTH-1 downto 0)	Imaginary component of sample.
SampleInReady	Output	std_logic	Nominal upstream-ready output; no acceptance handshake.
SampleOutValid	Output	std_logic	High when output is valid.
SampleOutLast	Output	std_logic	Pass through.
OverflowStatus	Output	std_logic_vector(1 downto 0)	[1]: I [0]: Q
SampleOut_I	Output	signed(BIT_WIDTH-1 downto 0)	Real component of mixed sample.
SampleOut_Q	Output	signed(BIT_WIDTH-1 downto 0)	Imaginary component of mixed sample.

3.3.5 Low-Pass Filter

A finite impulse response (FIR) filter has a finite-duration impulse response. Filter coefficients are calculated from the desired properties of the filter in MATLAB or Python. An L -tap FIR implements the convolution sum

$$Y[n] = \sum_{i=0}^{L-1} b[i]X[n-i].$$

Here, $b[i]$ are the L coefficients, and $X[n]$ is the input sequence. A causal FIR with L taps has order $L-1$. This design uses the two-tap, first-order moving-average low-pass filter $b[0] = b[1] = 0.5$.

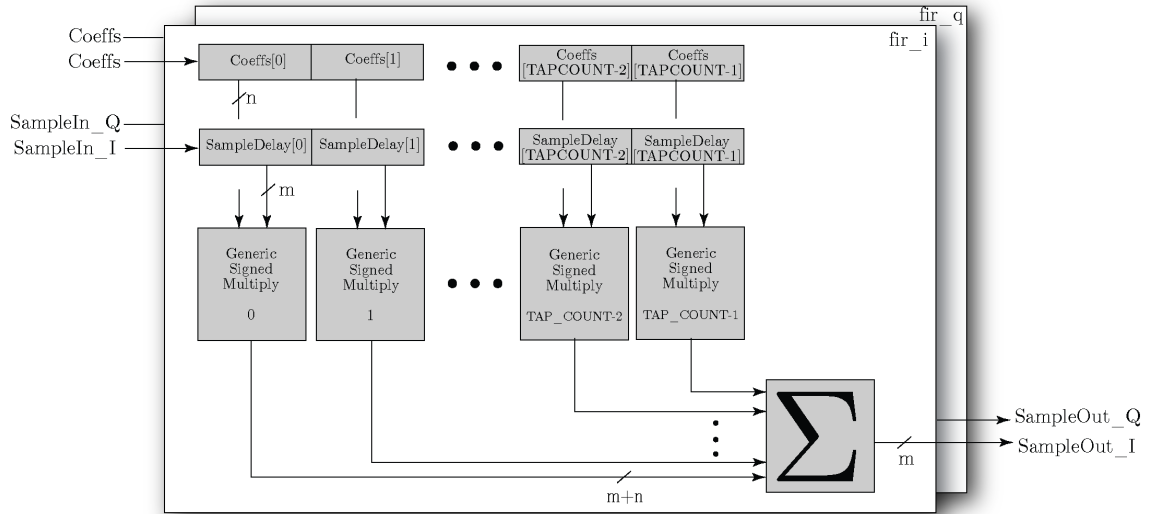


Figure 25: Digital design of the LPF module.

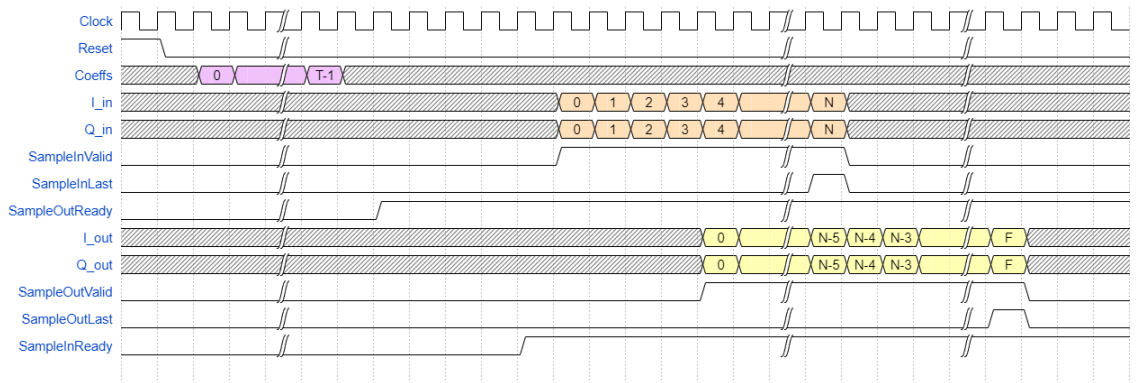


Figure 26: Timing diagram of the low-pass filter module.

Table V: Configured low-pass-filter interface.

Port Name	Direction	Data Type	Description
TAP_COUNT	Config.	natural	Number of coefficients.
TAP_WIDTH	Config.	natural	Coefficient width: one sign bit and TAP_WIDTH-1 fractional bits.
BIT_WIDTH	Config.	natural	Sample width: one sign bit and BIT_WIDTH-1 fractional bits.
Clock	Input	std_logic	Synchronous clock.
Reset	Input	std_logic	Synchronous, active-high reset.
Coeffs	Input	CoeffArray(TAP_COUNT-1 downto 0)	TAP_COUNT signed coefficients, with $b[0] = b[1] = 0.5$ in this application.
SampleOutReady	Input	std_logic	Nominal downstream-ready input; does not stall.
SampleInValid	Input	std_logic	High when input is valid.
SampleInLast	Input	std_logic	Pass through.
SampleIn_I	Input	signed(BIT_WIDTH-1 downto 0)	Real component of sample.
SampleIn_Q	Input	signed(BIT_WIDTH-1 downto 0)	Imaginary component of sample.
SampleInReady	Output	std_logic	Nominal upstream-ready output; no acceptance handshake.
SampleOutValid	Output	std_logic	High when output is valid.
SampleOutLast	Output	std_logic	Pass through.
SampleOut_I	Output	signed(BIT_WIDTH-1 downto 0)	Real component of filtered sample.
SampleOut_Q	Output	signed(BIT_WIDTH-1 downto 0)	Imaginary component of filtered sample.

3.3.6 Instantaneous Power

This module computes the pointwise magnitude squared of a complex sample. It supplies the $\tau = 0$ product $X[n]X^*[n]$ used by the zero-lag AF and CAF:

$$\begin{aligned} P_X[n] &= X[n]X^*[n] \\ &= |X[n]|^2 \\ &= \Re\{X[n]\}^2 + \Im\{X[n]\}^2. \end{aligned}$$

The real and imaginary components are squared in parallel and then added. This pointwise identity is not Parseval's theorem, which instead relates a sum or integral of squared magnitudes in the time domain to one in the frequency domain. For compactness, the following arithmetic-module tables use the documentation labels LEN_A, LEN_B, TRUNC, and RND for the VHDL generics `LengthA`, `LengthB`, `BitstoTrunc`, and `BitstoRound`, respectively.

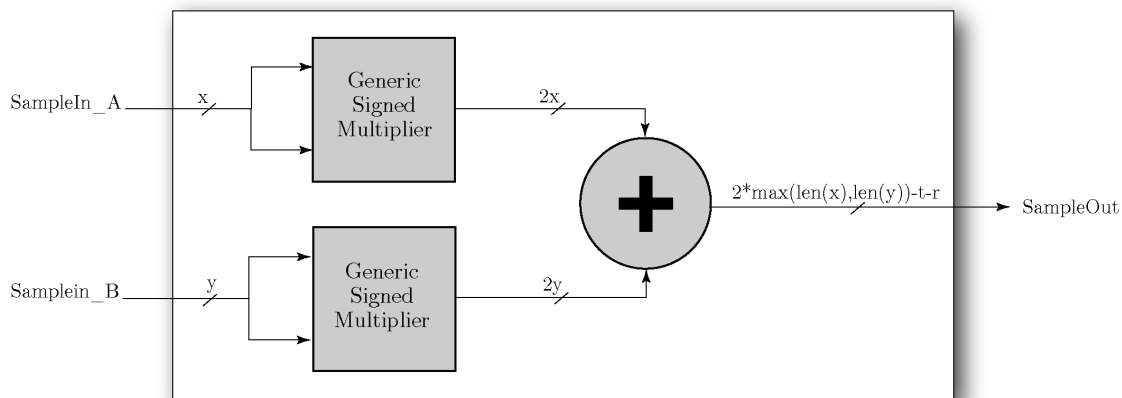


Figure 27: Digital design of the instantaneous power module.

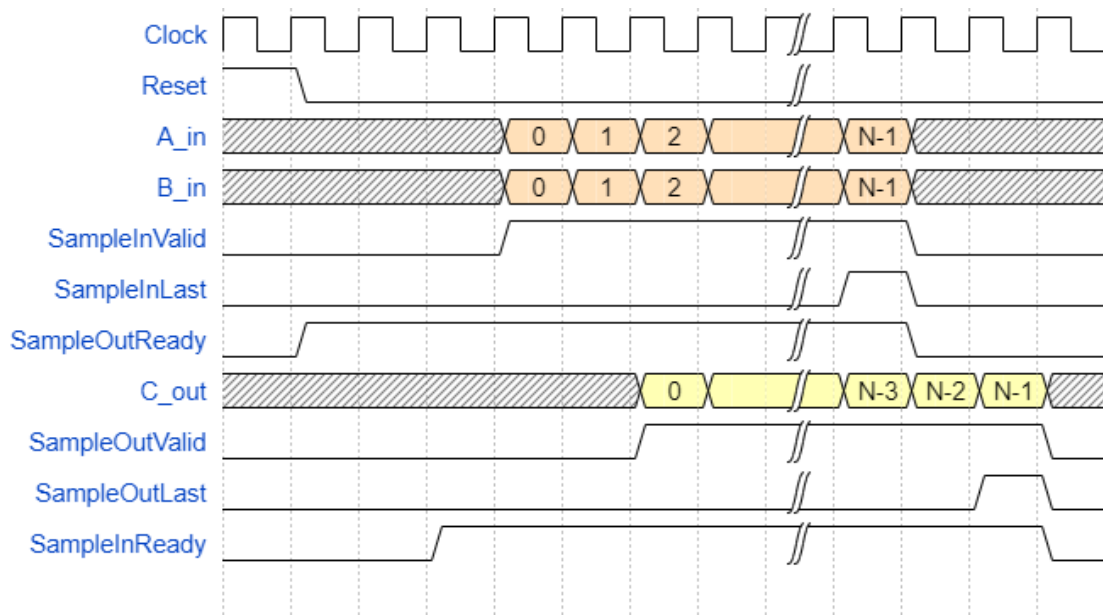


Figure 28: Timing diagram of the instantaneous power module.

Table VI: Configured instantaneous-power interface.

Port Name	Direction	Data Type	Description
LEN_A	Generic	natural	Bit width of the real input.
LEN_B	Generic	natural	Bit width of the imaginary input.
LATENCY	Generic	natural	Controls the total operation latency.
TRUNC	Generic	natural	Number of most-significant bits truncated.
RND	Generic	natural	Number of least-significant bits removed after rounding.
Clock	Input	std_logic	Synchronous clock.
Reset	Input	std_logic	Synchronous, active-high reset.
SampleOutReady	Input	std_logic	Nominal downstream-ready input; does not stall.
SampleInValid	Input	std_logic	High when input is valid.
SampleInLast	Input	std_logic	Pass through.
SampleIn_A	Input	signed(LEN_A-1 downto 0)	Real component of sample.
SampleIn_B	Input	signed(LEN_B-1 downto 0)	Imaginary component of sample.
SampleInReady	Output	std_logic	Nominal upstream-ready output; no acceptance handshake.
SampleOutValid	Output	std_logic	High when output is valid.
SampleOutLast	Output	std_logic	Pass through.
SampleOut	Output	unsigned(2*max(LEN_A, LEN_B)-1-RND-TRUNC downto 0)	Nonnegative magnitude-squared output.

3.3.7 Complex Multiplier

The product of two complex numbers can be expressed in terms of pairs of real

and imaginary components as follows:

$$X_1[n]X_2[n] = (A + jB)(C + jD)$$

$$\Re\{X_1[n]X_2[n]\} = AC - BD$$

$$\Im\{X_1[n]X_2[n]\} = AD + BC.$$

Here, the multiplication of complex values $X_1[n]X_2[n]$ simplifies to four real multiplications, one real addition, and one real subtraction.

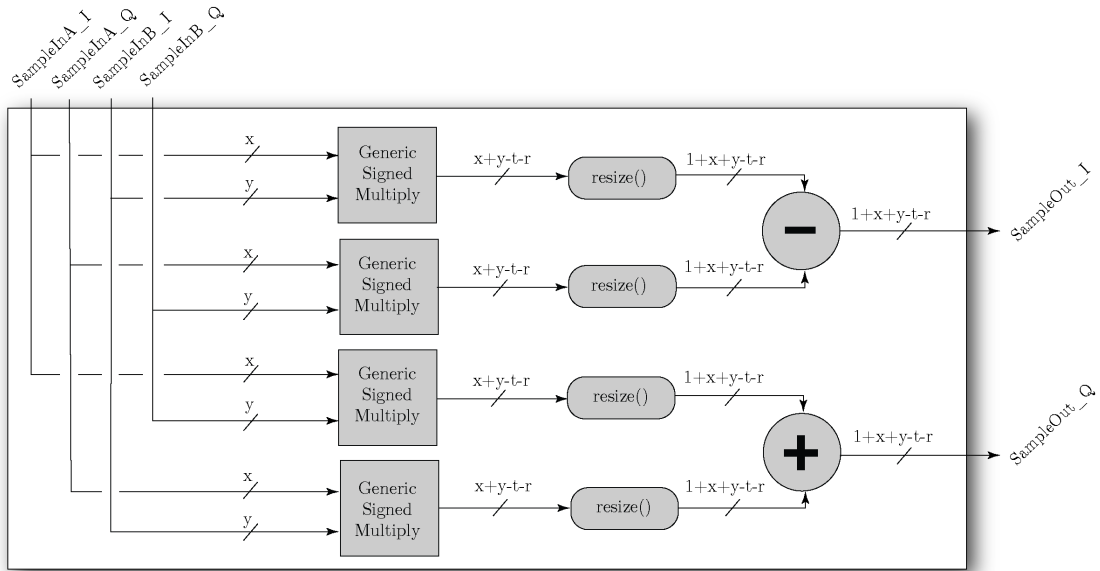


Figure 29: Digital design of the complex multiplier module.

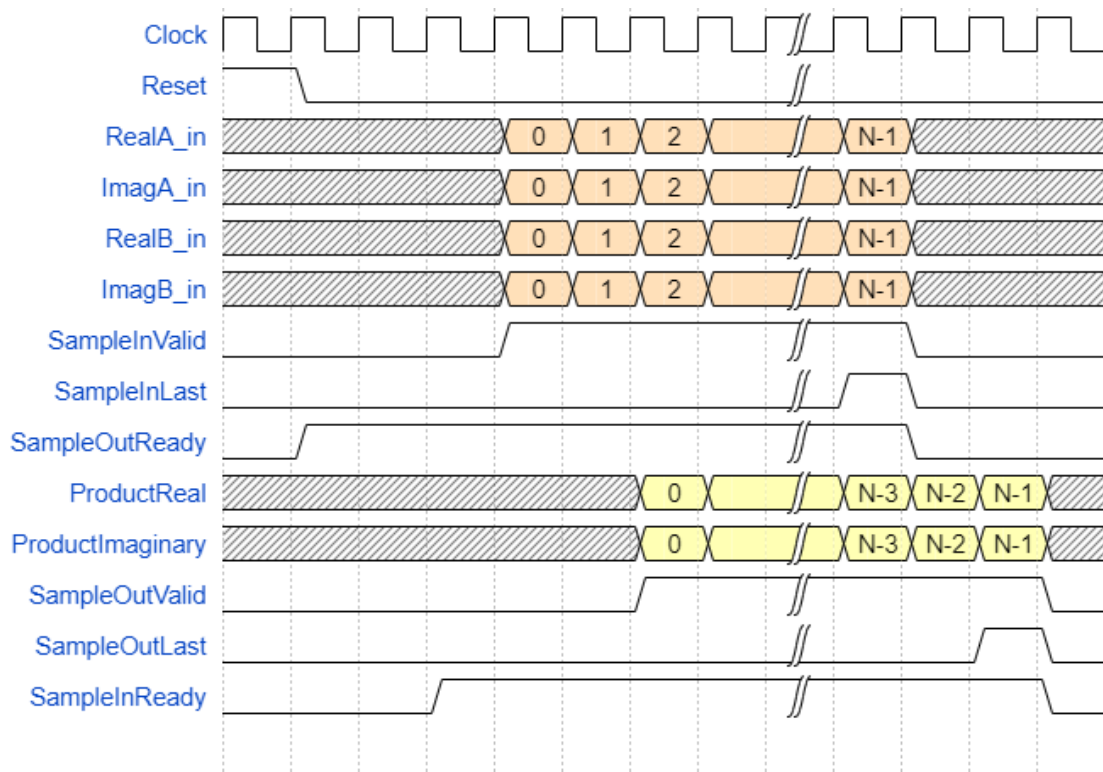


Figure 30: Timing diagram of the complex multiplier module.

Table VII: Configured complex-multiplier interface.

Port Name	Direction	Data Type	Description
LEN_A	Generic	natural	Operand-A component width.
LEN_B	Generic	natural	Operand-B component width.
LATENCY	Generic	natural	Total operation latency.
TRUNC	Generic	natural	Most-significant bits truncated.
RND	Generic	natural	Least-significant bits removed after rounding.
Clock	Input	std_logic	Synchronous clock.
Reset	Input	std_logic	Active-high synchronous reset.
SampleOutReady	Input	std_logic	Nominal downstream-ready input; does not stall.
SampleInValid	Input	std_logic	Input is valid.
SampleInLast	Input	std_logic	Pass through.
SampleIn_I_A	Input	signed(LEN_A-1 downto 0)	Operand-A real component.
SampleIn_Q_A	Input	signed(LEN_A-1 downto 0)	Operand-A imaginary component.
SampleIn_I_B	Input	signed(LEN_B-1 downto 0)	Operand-B real component.
SampleIn_Q_B	Input	signed(LEN_B-1 downto 0)	Operand-B imaginary component.
SampleInReady	Output	std_logic	Nominal upstream-ready output; no acceptance handshake.
SampleOutValid	Output	std_logic	Output is valid.
SampleOutLast	Output	std_logic	Pass through.
SampleOut_I	Output	signed(LEN_A+LEN_B-RND-TRUNC downto 0)	Real component of product.
SampleOut_Q	Output	signed(LEN_A+LEN_B-RND-TRUNC downto 0)	Imaginary component of product.

3.3.8 Multiplier

There are many real multipliers used throughout this design. Although multiplication is an HDL operation, the module provides a fixed-latency pipeline together with explicit bit-width, truncation, saturation, and rounding controls. Its ready/-valid signals delay interface metadata but do not stall the arithmetic pipeline, so it is not a backpressure-capable handshake implementation. Mathematically, this module performs the multiplication of two real values, expressed as:

$$Y[n] = X_1[n]X_2[n]. \quad (28)$$

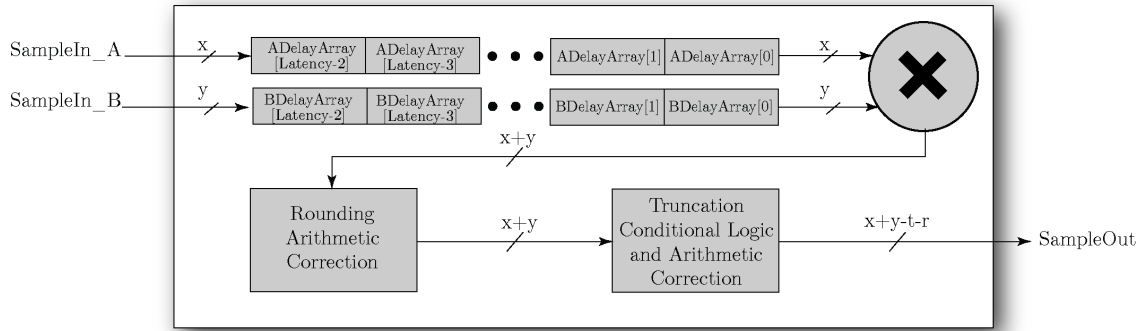


Figure 31: Digital design of the multiplier module.

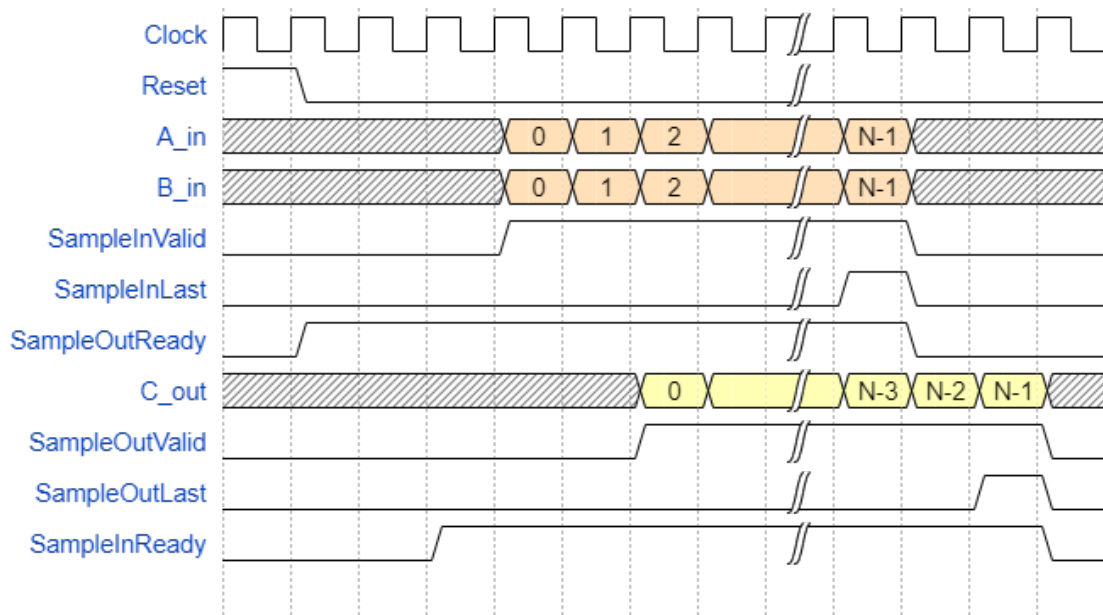


Figure 32: Timing diagram of the multiplier module.

Table VIII: Configured multiplier interface.

Port Name	Direction	Data Type	Description
LEN_A	Generic	natural	Bit width of operand A.
LEN_B	Generic	natural	Bit width of operand B.
LATENCY	Generic	natural	Controls the total operation latency.
TRUNC	Generic	natural	Number of most-significant bits truncated.
RND	Generic	natural	Number of least-significant bits removed after rounding.
Clock	Input	std_logic	Synchronous clock.
Reset	Input	std_logic	Synchronous, active-high reset.
SampleOutReady	Input	std_logic	Nominal downstream-ready input; does not stall.
SampleInValid	Input	std_logic	High when input is valid.
SampleInLast	Input	std_logic	Pass through.
SampleIn_A	Input	signed(LEN_A-1 downto 0)	Real-valued operand A.
SampleIn_B	Input	signed(LEN_B-1 downto 0)	Real-valued operand B.
SampleInReady	Output	std_logic	Nominal upstream-ready output; no acceptance handshake.
SampleOutValid	Output	std_logic	High when output is valid.
SampleOutLast	Output	std_logic	Pass through.
SampleOut	Output	signed(LEN_A+LEN_B-1-RND-TRUNC downto 0)	Real-valued product.

3.3.9 Accumulator

The accumulator implements the running sum used by each single-frequency Fourier evaluation and participates in the integrate-and-dump flow control between

sample sets. For a sequence whose first accepted sample has local index $n = 0$, the intended recurrence is

$$S[n] = \begin{cases} X[n], & n = 0, \\ S[n - 1] + X[n], & n > 0 \end{cases}. \quad (29)$$

Here, $X[n]$ is an accepted input sample and $S[n]$ is the running sum within the current sequence. The module itself is real-valued; the DFT instantiates separate accumulators for the in-phase and quadrature components. The VHDL presents a fixed-width slice of its wider internal sum, so this step performs fixed-point scaling and truncation rather than an exact arithmetic division by N . With the default 28-bit sum and 16-bit output, `Sum(27 downto 12)` is effectively a 12-bit right shift; for $N = 2^{16}$, it is not the 16-bit shift corresponding to division by N .

The archived RTL and testbench do not implement the intended boundary recurrence exactly. In the RTL, `N_SAMPLES` is unused and the internal width `NORM_FACTOR` is fixed at 28 bits. On a clock for which `SampleInLast` is asserted, `SampleOut` receives the pre-edge sum while the internal sum is loaded with the marked input. Because the archived testbench asserts `SampleInLast` on sample $N - 1$, the emitted closing sum excludes the marked sample, which instead seeds the next block. Moreover, an overflow-free sum of 2^{16} arbitrary 16-bit signed inputs requires $16 + \log_2(2^{16}) = 32$ bits in the worst case, so the fixed 28-bit internal width cannot guarantee overflow-free accumulation. When `SampleInvalid` is low, the process leaves `SampleInReady`, `SampleOutValid`, and `SampleOutLast` unchanged, so asserted status can persist rather than forming one-cycle handshake events. The boundary, scaling, width, and interface-control logic therefore require correction and re-verification before the RTL is reused.

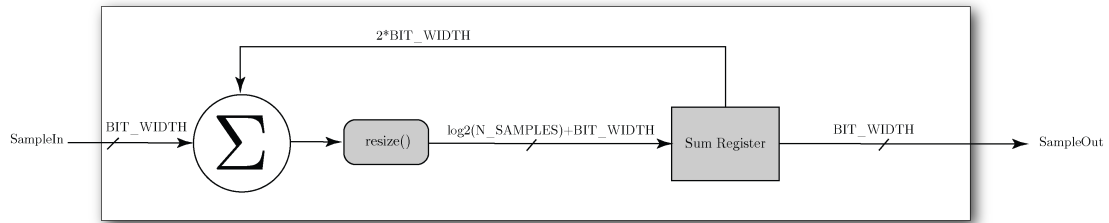


Figure 33: Digital design of the accumulator module.

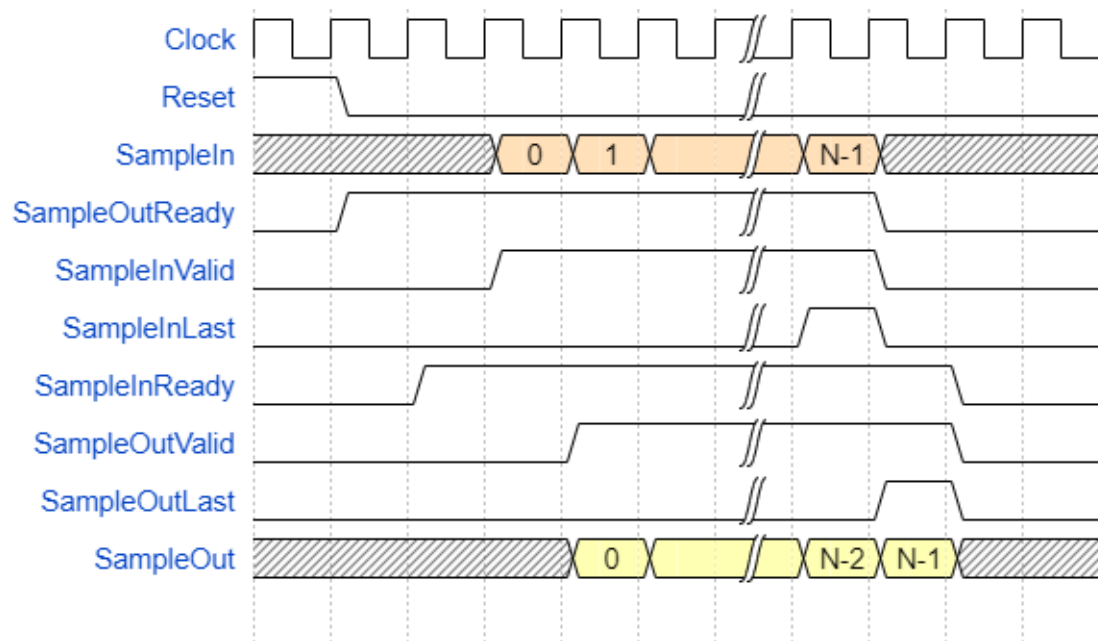


Figure 34: Intended timing of the accumulator module; the archived RTL differs as described in the text.

Table IX: Configured accumulator interface.

Port Name	Direction	Data Type	Description
N_SAMPLES	Generic	natural	Requested length; unused by the archived RTL.
BIT_WIDTH	Generic	natural	Sample width: one sign bit and BIT_WIDTH-1 fractional bits.
Clock	Input	std_logic	Synchronous clock.
Reset	Input	std_logic	Synchronous, active-high reset.
SampleOutReady	Input	std_logic	Nominal downstream-ready input; does not stall.
SampleInValid	Input	std_logic	High when input is valid.
SampleInLast	Input	std_logic	Marks a sequence boundary.
SampleIn	Input	signed(BIT_WIDTH-1 downto 0)	Real-valued input sample.
SampleInReady	Output	std_logic	Nominal upstream-ready output; no acceptance handshake.
SampleOutValid	Output	std_logic	High when output is valid.
SampleOutLast	Output	std_logic	Propagated sequence-boundary marker.
SampleOut	Output	signed(BIT_WIDTH-1 downto 0)	Scaled running-sum output.

CHAPTER IV

PERFORMANCE CHARACTERIZATION

This chapter characterizes the streaming SCA’s ability to sense and estimate the CF and SR of signals in a monitored spectrum. First, a Python model generates an M -QAM-SRRC waveform, applies the software channel, quantizes the in-phase and quadrature (IQ) samples, and saves them to a file. A VHDL testbench reads that file, evaluates the channelized zero-lag CAF statistic $T_y(v, k) = |\hat{C}_y(\alpha_v, f_k)|^2$ over user-defined requested SR and CF candidate grids $\mathcal{R}$ and $\mathcal{F}$, and saves the results. MATLAB then plots the statistic and the constant threshold $\gamma = \beta \hat{P}_y^2$. For each rate candidate $\alpha_v \in \mathcal{R}$, the CF candidate is the $f_k \in \mathcal{F}$ having the largest statistic, and that row exceeds the threshold only when its maximum is greater than γ . Table XII gives a corrected reconstruction of the selected peak pair or pairs: legacy normalized plot coordinates are converted to nominal physical units, and entries inconsistent with the plots and experiment definitions are corrected. It does not list every above-threshold row. Thus, v and k are candidate-array indices, not physical frequencies or automatically detected bins.

4.1 Experiments

Characterization is performed by first establishing baseline parameters for the transmitter and receiver. Although the baseline parameters yield a clear threshold crossing in the archived single-trial result, they may not reflect practical constraints

such as hardware restrictions or channel impairments. Each subsequent experiment varies one listed parameter relative to the baseline. The parameters evaluated are shown in Table X and Table XI.

Table X: Python transmitter and channel.

Here, $F_s = 1$ MHz except in Experiment VII, where $F_s = 10$ MHz; r is the roll-off factor. In the two-signal Experiments XII and XIII, the scalar E_b/N_0 applies to both components.

Expt.	M	r	R_s (MBd)	f_c (MHz)	$\frac{E_b}{N_0}$ (dB)
I	4	0.35	0.10	0.05	14.00
II	16	0.35	0.10	0.05	14.00
III	4	0.20	0.10	0.05	14.00
IV	4	0.35	0.01	0.05	14.00
V	4	0.35	0.10	−0.05	14.00
VI	4	0.35	0.10	0.05	9.00
VII	4	0.35	0.10	0.05	14.00
VIII	4	0.35	0.10	0.05	14.00
IX	4	0.35	0.10	0.05	14.00
X	4	0.35	0.10	0.05	14.00
XI	4	0.35	0.10	0.05	14.00
XII	4; 16	0.35; 0.35	0.10; 0.25	−0.20; 0.20	14.00
XIII	4; 16	0.35; 0.35	0.10; 0.25	0.05; 0.10	14.00

Table XI: VHDL receiver.

Here, BW is the bit width, and $\mathcal{R}$ and $\mathcal{F}$ are inclusive requested (nominal) physical search grids. The notation $[a : s : b]$ means values from a through b in steps of s .

Expt.	F_s (MHz)	N	BW	$\mathcal{R}$ (MBd)	$\mathcal{F}$ (MHz)
I	1.00	2^{16}	16	$[0.10 : 0.10 : 0.50]$	$[-0.50 : 0.05 : 0.50]$
II	1.00	2^{16}	16	$[0.10 : 0.10 : 0.50]$	$[-0.50 : 0.05 : 0.50]$
III	1.00	2^{16}	16	$[0.10 : 0.10 : 0.50]$	$[-0.50 : 0.05 : 0.50]$
IV	1.00	2^{16}	16	$[0.01 : 0.01 : 0.50]$	$[-0.50 : 0.05 : 0.50]$
V	1.00	2^{16}	16	$[0.10 : 0.10 : 0.50]$	$[-0.50 : 0.05 : 0.50]$
VI	1.00	2^{16}	16	$[0.10 : 0.10 : 0.50]$	$[-0.50 : 0.05 : 0.50]$
VII	10.00	2^{16}	16	$[1.00 : 1.00 : 5.00]$	$[-5.00 : 0.05 : 5.00]$
VIII	1.00	2^{12}	16	$[0.10 : 0.10 : 0.50]$	$[-0.50 : 0.05 : 0.50]$
IX	1.00	2^{16}	8	$[0.10 : 0.10 : 0.50]$	$[-0.50 : 0.05 : 0.50]$
X	1.00	2^{16}	16	$\{0.10\}$	$[-0.50 : 0.01 : 0.50]$
XI	1.00	2^{16}	16	$[0.0999 : 10^{-6} : 0.1001]$	$\{0.05\}$
XII	1.00	2^{16}	16	$[0.05 : 0.05 : 0.50]$	$[-0.50 : 0.05 : 0.50]$
XIII	1.00	2^{16}	16	$[0.05 : 0.05 : 0.50]$	$[-0.50 : 0.05 : 0.50]$

The legacy plots display the requested physical candidates normalized by the sample rate: their horizontal coordinates are α_v/F_s and f_k/F_s in cycles per sample. Although the axes are labeled v and k , those plotted coordinates are not the one-based candidate-array indices defined in Chapter III. Likewise, the black curves are the nonnegative test statistic $T_y(v, k) = |\hat{C}_y(\alpha_v, f_k)|^2$ even though the legacy legends label them as $S_x^v[k]$. The requested CF grids reproduce both Nyquist endpoints, but $-F_s/2$ and $+F_s/2$ are the same discrete-time frequency and map to the same signed control word; consequently, those two requests are not distinct oscillator settings.

Experiment XI requested 201 symbol-rate values at 1-Hz intervals, but the 16-

bit control word at $F_s = 1$ MHz has a frequency quantum of $F_s/2^{16} = 15.2588$ Hz. Nearest-word quantization maps the requests to only 14 distinct settings, so the plot does not demonstrate 1-Hz hardware resolution.

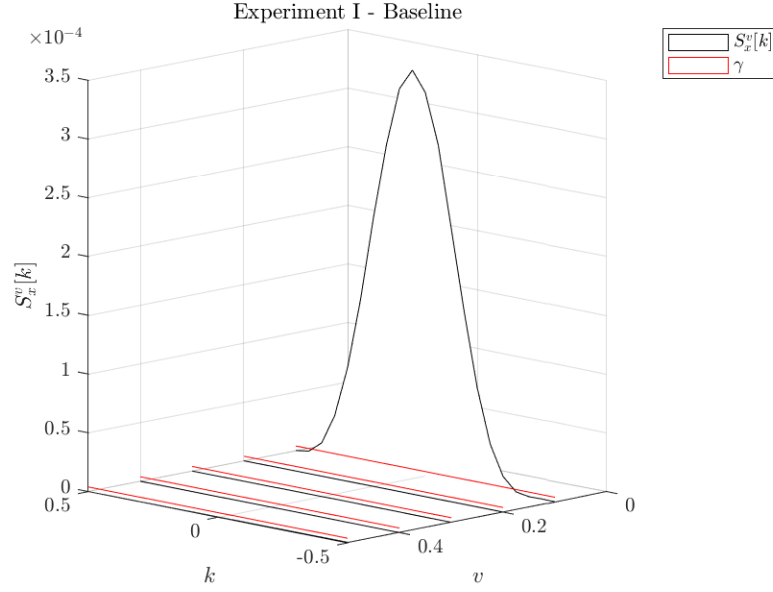


Figure 35: Streaming-SCA test-statistic plot for Experiment I.

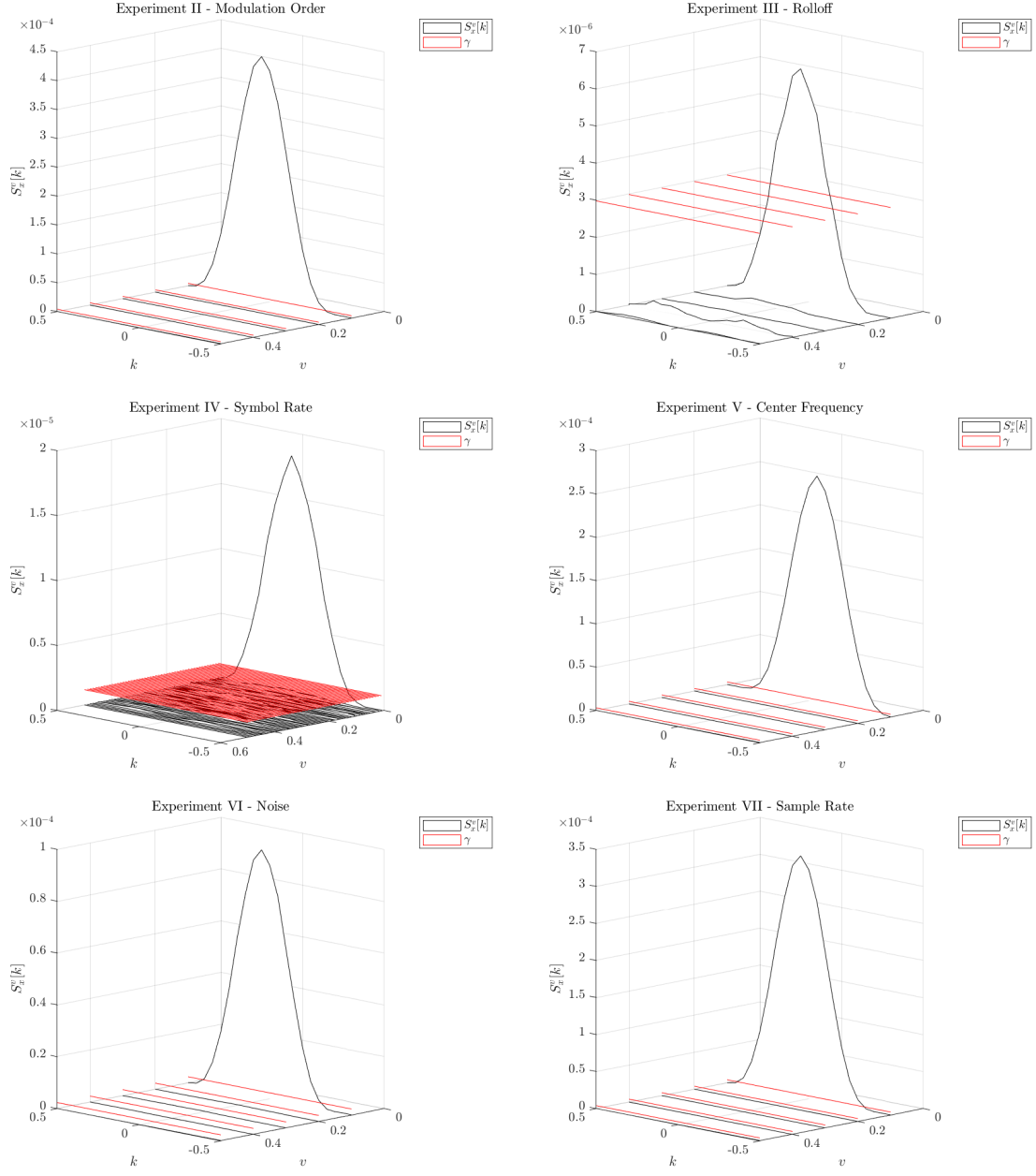


Figure 36: Streaming-SCA test-statistic plots for Experiments II–VII.

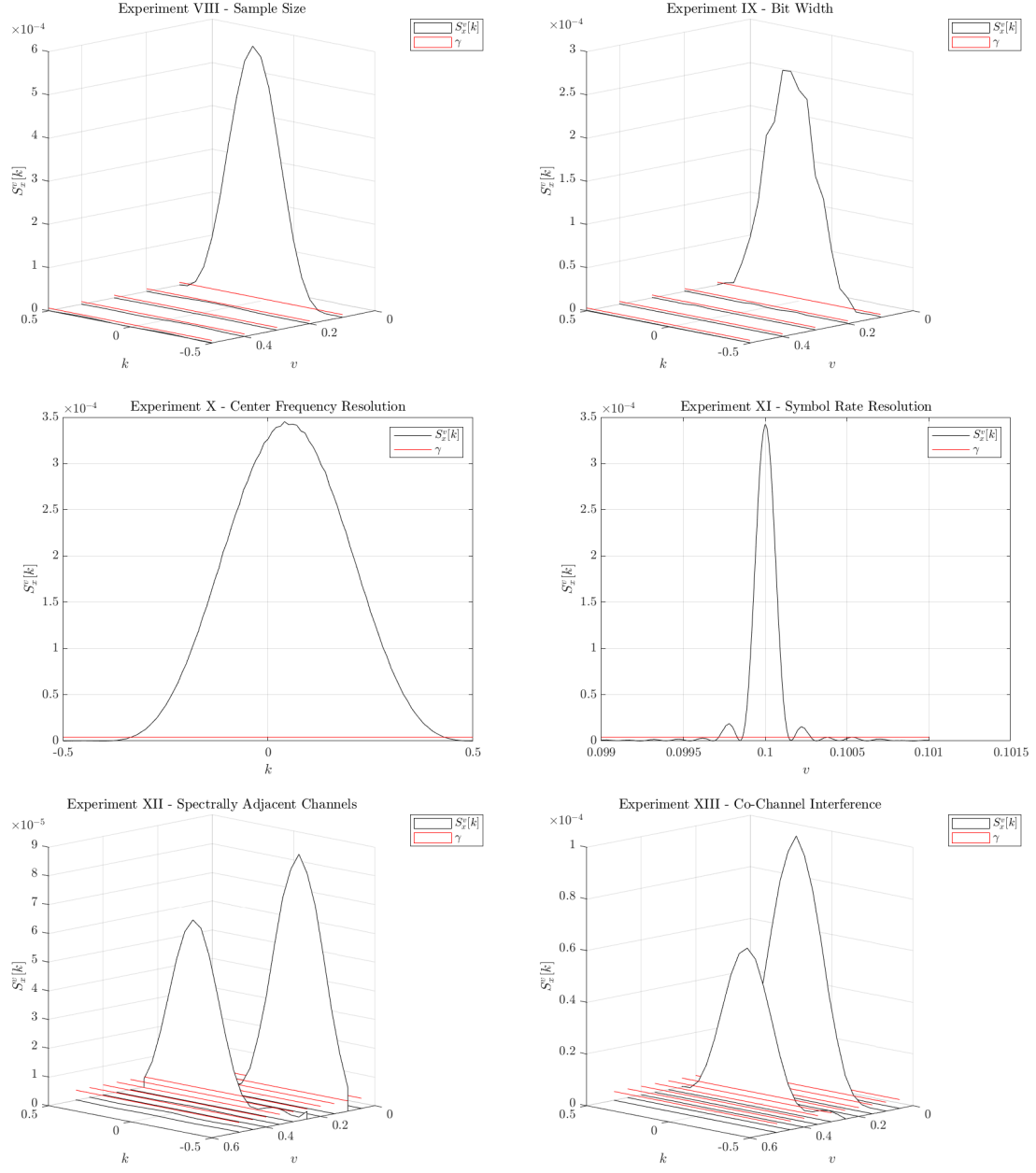


Figure 37: Streaming-SCA test-statistic plots for Experiments VIII–XIII.

4.2 Results

The corrected reconstruction of the archived results is summarized in Table XII, and Table XIII reproduces the reported resource counts. Every single-trial plot contains at least one statistic above the selected threshold, but a threshold crossing does not establish a probability of detection or guarantee correct parameter recovery. After conversion from the normalized plot coordinates to nominal physical units, eleven experiments report the transmitted pair or pairs. The reconstruction corrects Experiment IV’s rate to 0.01 MBd and Experiment XIII’s second CF to 0.10 MHz, matching the experiment definitions and plotted peaks. Experiment VII reports $R_s = 1.00$ MBd for a 0.10 MBd input because the receiver’s coarsest rate candidate was 1.00 MBd. Experiment X reports $f_c = 0.04$ MHz, one 10-kHz search step below the transmitted 0.05 MHz. The exact thesis-era RTL revision and the top-level synthesis, utilization, and timing reports were not retained. The recovered 2022 checkout contains the accumulator, power-format, and flow-control discrepancies documented in Chapter III, so it cannot independently establish that these archived simulations are bit-true to the ideal statistic.

The recovered component design reports and project-recreation scripts explicitly associate their module-level resource tables with a ZedBoard XC7Z020CLG484-1 and Vivado 2019.2.1 using the default 2019 synthesis and implementation strategies. They do not link the experiment-level counts in Table XIII to that setup. Metadata elsewhere in the later archive conflicts: most component source headers name an XC7Z045, later top-level and threshold headers name an XCKU025, and an unrelated out-of-context FFT checkpoint names an XCKU5P. Therefore the target, exact settings, and BRAM primitive-count convention behind Table XIII remain unresolved.

At one valid sample per clock, the nominal sample-processing time of a serial sweep is $N|\mathcal{R}||\mathcal{F}|/F_s$, excluding pipeline and control overhead. The raw timing logs are unavailable, so most listed delays remain unverified archival values; the retained values

do not fall below the calculated lower bound. A retained CFE testbench nevertheless provides one independent cross-check: its configuration matches Experiment VIII with $N = 4096$, five SR candidates, 21 CF candidates, and a 1-MHz sample clock. The resulting sample-processing lower bound is 0.430080 s, only $161\ \mu\text{s}$ below the reported 0.430241 s and thus consistent with finite pipeline and control overhead. Experiment VII's reported delay duplicates the baseline value despite its different sample rate and grid and is therefore anomalous. Experiment IV's copied value also equals the baseline time even though its 50-rate grid is ten times larger; its nominal sample-processing time alone is 68.8128 s, so Table XII omits the unavailable delay.

Table XII: Corrected summary of archived performance results.

Here, $\gamma = \beta \hat{P}_y^2$ with $\beta = 10^{-4}$; $\hat{P}_y^2$ is dimensionless after input normalization and is rounded to six significant digits. Each tuple is one reconstructed nominal (R_s, f_c) pair in (MBd, MHz).

Expt.	Reported delay (s)	$\hat{P}_y^2$	Nominal pair(s)
I	6.881441	0.0390606	(0.10, 0.05)
II	6.881441	0.0398833	(0.10, 0.05)
III	6.881441	0.0298027	(0.10, 0.05)
IV	—	0.0117707	(0.01, 0.05)
V	6.881441	0.0296905	(0.10, -0.05)
VI	6.881441	0.0230728	(0.10, 0.05)
VII	6.881441	0.0394264	(1.00, 0.05)
VIII	0.430241	0.0613674	(0.10, 0.05)
IX	6.881441	0.0323822	(0.10, 0.05)
X	6.619201	0.0390606	(0.10, 0.04)
XI	13.243154	0.0390606	(0.10, 0.05)
XII	13.762841	0.0353092	(0.10, -0.20); (0.25, 0.20)
XIII	13.762841	0.0341933	(0.10, 0.05); (0.25, 0.10)

Table XIII: Archived experiment-level resource counts; target and settings are unresolved.

Expt.	LUT	LUTRAM	FF	BRAM	DSP	I/O
I	814	3	391	15	16	108
II	814	3	391	15	16	108
III	814	3	391	15	16	108
IV	814	3	391	15	16	108
V	814	3	391	15	16	108
VI	814	3	391	15	16	108
VII	814	3	391	15	16	108
VIII	814	3	391	15	16	108
IX	674	3	319	15	16	100
X	814	3	391	15	16	108
XI	814	3	391	15	16	108
XII	814	3	391	15	16	108
XIII	814	3	391	15	16	108

CHAPTER V

CONCLUSION

We investigated a low-memory, FPGA-targeted VHDL architecture for M -QAM-SRRC SR-CF detection. We began by defining the streaming SCA’s scope and then described its sample-by-sample datapath. The datapath does not instantiate an FFT or vendor-specific FFT intellectual-property core, although MATLAB and the Vivado toolchain were used for modeling, simulation, and synthesis. Finally, we presented thirteen single-trial VHDL simulations and the archived resource counts. Recovered 2021 component design reports and project scripts identify the intended development platform as a ZedBoard XC7Z020CLG484-1 with Vivado 2019.2.1. However, the exact thesis-era RTL revision and top-level synthesis reports are unavailable, later target metadata conflicts, and the recovered 2022 checkout contains the fixed-point and flow-control discrepancies documented in Chapter III. The archived simulations therefore cannot establish bit-true conformance, and the experiment-level resource counts cannot be independently tied to the documented component setup. None of these results are measurements from deployed flight hardware.

As discussed in Section 3.2, one shortcoming of this design is that its serial processing time grows as the SR and CF candidate grids become finer. Koch first forms spectral segments from a thresholded Welch PSD [5]. After centering and dynamically low-pass filtering each segment, the cyclic-autocorrelation FFT (CAC-FFT) forms $z[n] = |x[n]|^2$ and uses an N -point FFT to evaluate the zero-lag CAC on a

uniformly spaced SR grid; peak finding selects candidate rates. For each selected rate, Koch’s implemented carrier-frequency detector then sweeps that segment across frequency offsets, applies the rate-dependent low-pass filter, evaluates a single-rate CAC, and thresholds the normalized peak. A separate center-frequency preprocessor in the report was preliminary and was not used. The hardware described in this thesis serializes and simplifies the carrier sweep, but it integrates neither the preceding PSD segmentation nor the CAC-FFT candidate-selection stage. Adding those stages is a possible extension; it would require additional filtering, block storage, and an FFT, and the reduction in serial work would depend on how many candidates survive preprocessing, so no fixed reduction factor is established here.

Decimation is another possible way to reduce the sample rate processed by a candidate branch, but the factor is not generally F_s/R_s . After the band of interest is centered, an anti-alias low-pass filter must precede integer downsampling by D , giving $F'_s = F_s/D$. The value of D must preserve the occupied bandwidth and the cyclic frequencies needed by the detector under the new Nyquist limit. It also changes the NCO control-word mapping, the observation interval represented by N samples, and the scheduling of parallel streaming-SCA instances. This thesis does not implement or evaluate such a decimator.

The constant $\beta = 10^{-4}$ used in the characterization was selected empirically; the thirteen single-trial plots do not calibrate a false-alarm probability or establish a probability of detection. A recovered exploratory ROC notebook supplies no additional evidence because its stored run terminates with an execution error rather than a completed sweep. The present row-maximum rule can report at most one CF for each SR candidate, so it cannot resolve two signals that share an SR but have different CFs. Detecting multiple local CF maxima above threshold, followed by suitable nonmaximum suppression or clustering, is one possible extension. Similarly, a fine SR grid can yield a cluster of correlated above-threshold candidates around one

physical feature. Those adjacent grid responses are not independent statistical false alarms and should be consolidated before a waveform count is reported. Selecting and validating such postprocessing requires Monte Carlo receiver-operating-characteristic analysis over the intended channels and signal classes.

Lastly, the characterization tests only 4-QAM-SRRC and 16-QAM-SRRC signals. The same datapath can evaluate its statistic for other sampled waveforms, but that fact alone does not establish useful cyclic features or reliable detection. Rectangular pulse shapes, FSK, OOK, MSK, DSSS, DPSK, and Manchester coding were not tested here; their candidate grids, filtering, and thresholds would have to be derived and validated before support could be claimed.

REFERENCES

- [1] S. Bandyopadhyay, G. P. Subramanian, R. Foust, D. Morgan, S.-J. Chung, and F. Y. Hadaegh. A review of impending small satellite formation flying missions. In *Proceedings of the 53rd AIAA Aerospace Sciences Meeting*, pages 1–17, Kissimmee, Florida, 2015. American Institute of Aeronautics and Astronautics. doi: 10.2514/6.2015-1623.
- [2] W. A. Gardner. *Introduction to Random Processes with Applications to Signals and Systems*. McGraw-Hill, New York City, New York, 2 edition, 1990.
- [3] W. A. Gardner, A. Napolitano, and L. Paura. Cyclostationarity: Half a century of research. *Signal Processing*, 86(4):639–697, 2006. doi: 10.1016/j.sigpro.2005.06.016.
- [4] I. Ilyas, S. Paul, A. Rahman, and R. K. Kundu. Comparative evaluation of cyclostationary detection based cognitive spectrum sensing. In *2016 IEEE 7th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*, pages 1–7, New York City, New York, 2016. IEEE. doi: 10.1109/UEMCON.2016.7777887.
- [5] M. V. Koch and J. A. Downey. Interference mitigation using cyclic autocorrelation and multi-objective optimization. Technical Report NASA/TM–2019-220226, National Aeronautics and Space Administration, Glenn Research Center, Cleveland, Ohio, July 2019.
- [6] Y. C. Liang. Spectrum sensing theories and methods. In *Dynamic Spectrum Management: From Cognitive Radio to Blockchain and Artificial Intelligence*, pages 41–85. Springer, Singapore, 2020. doi: 10.1007/978-981-15-0776-2_3.
- [7] C. M. Spooner. *Theory and Application of Higher-Order Cyclostationarity*. PhD thesis, University of California, Davis, Davis, California, June 1992.

- [8] H. L. Van Trees. Classical detection and estimation theory. In *Detection, Estimation, and Modulation Theory, Part I*, pages 19–165. Wiley-Interscience, Hoboken, NJ, 2001. doi: 10.1002/0471221082.ch2.
- [9] Xilinx, Inc. Zynq-7000 SoC Data Sheet: Overview. Technical Report DS190, Xilinx, Inc., July 2018. Version 1.11.1.

APPENDIX

APPENDIX A

SOURCE CODE

The following listing is imported directly from the script used to generate Figure 16, so the appendix and the executable model remain synchronized. The model uses one receiver-noise realization for both signals, applies a common AGC scale factor, and compares $T_y(v, k) = |\hat{C}_y(\alpha_v, f_k)|^2$ with the constant threshold $\gamma = \beta \hat{P}_y^2$.

```
%%
% Dylan J. Gormley (NASA GRC/LCI) - 2020

clear
close all
rng(20210901, 'twister')
%%
% Transmitter: M-QAM-SRRC

Nmessages = 2^16; % number of transmitted symbols per component
Fs         = 1.0e6; % sample rate

Rs         = [0.10 0.20]*1e6; % symbol rate
fc         = [0.05 0.10]*1e6; % carrier frequency
M          = [4      16];      % modulation order
Rolloff    = [1.00 0.35];      % excess bandwidth
EbNO       = [14.0  9.0];      % energy per bit over noise density in dB

% Extract clean signals. A single shared receiver noise floor is
% added
% below; independent transmitter noise realizations must not be
% stacked.
[~, syms_pb1_clean] = qam_srrc_xmtr(Nmessages, Fs, Rs(1), fc(1),
    M(1), Rolloff(1), EbNO(1));
[~, syms_pb2_clean] = qam_srrc_xmtr(Nmessages, Fs, Rs(2), fc(2),
    M(2), Rolloff(2), EbNO(2));

%%
% RX RF front end

% Sample size used to compute one point.
Ncapture = 2^16;
components = [syms_pb1_clean(1:Ncapture), syms_pb2_clean(1:Ncapture)];
```

```

% Realize both requested Eb/NO values against one physical noise
  floor.
% Keep signal 1 at its generated level and scale signal 2
  accordingly.
SPS      = Fs./Rs;
CNR_target = EbNO + 10*log10(log2(M)) - 10*log10(SPS);
component_pwr = mean(abs(components).^2,1);
noise_pwr    = component_pwr(1)/10^(CNR_target(1)/10);
target_pwr   = noise_pwr*10.^(CNR_target/10);
components   = components.*sqrt(target_pwr./component_pwr);

unit_noise = (randn(Ncapture,1) + 1j*randn(Ncapture,1))/sqrt(2);
noise      = unit_noise*sqrt(noise_pwr/mean(abs(unit_noise).^2));
syms_rx    = sum(components,2) + noise;

% Simulate receiver AGC. Its common scale factor preserves both
  CNRs.
norm_factor = modnorm(syms_rx,'peakpow',1);
agc         = syms_rx*norm_factor;

measured_CNR =
    10*log10(mean(abs(components).^2,1)/mean(abs(noise).^2));
measured_EbNO = measured_CNR - 10*log10(log2(M)) + 10*log10(SPS);
assert(all(abs(measured_EbNO-EbNO) < 0.05), ...
    'A realized Eb/NO differs from its request by at least 0.05 dB. ');
fprintf('Measured Eb/NO: %.2f dB, %.2f dB\n',measured_EbNO);

%%
% Receiver: streaming SCA (fully vectorized)

% Valid symbol-rate range is (0, Fs/2].
a_step = min(Rs);
a_array = (a_step:a_step:FsWithout/2)';

% The half-open 10 kHz grid includes both transmitted carrier
  frequencies
% exactly without duplicating the equivalent -Fs/2 and +Fs/2
  Nyquist LOs.
fc_step = 0.010e6;
fc_array = (-FsWithout/2:fc_step:FsWithout/2-fc_step)';

% Two-tap low-pass filter
hlpf = [0.5,0.5];

```

```

n = (0:Ncapture-1)';

% 1. Calculate frequency bins once
vbin = round(a_array/Fs*Ncapture) + 1;

% 2. Generate downconverting local oscillators. The transmitter
    uses the
% positive-frequency convention  $\exp(+j*2*\pi*fc*n/Fs)$ .
LO_matrix = exp(-1j*2*pi*n*(fc_array')/Fs);

% 3. Apply the local oscillators using implicit expansion
mixed_signal = agc(1:Ncapture) .* LO_matrix;

% 4. Channelize
channelized = filter(hlpf, 1, mixed_signal);

% 5. Symbol rate estimation
R = abs(channelized).^2;

% 6. Vectorized Goertzel algorithm
S = goertzel(R, vbin) / Ncapture;

% Test statistic and constant threshold.
T = abs(S).^2; % test statistic
beta      = 1e-3; % user-selected threshold scale
P_y_hat   = sum(abs(agc).^2)/Ncapture;
gamma      = beta*P_y_hat^2;
threshold = gamma*ones(size(T));

% Report the strongest carrier candidate at each transmitted symbol
    rate.
peak_T     = zeros(size(Rs));
peak_fc    = zeros(size(fc));
is_detected = false(size(Rs));
for idx = 1:numel(Rs)
    row = find(a_array == Rs(idx),1);
    [peak_T(idx),column] = max(T(row,:));
    peak_fc(idx) = fc_array(column);
    is_detected(idx) = peak_T(idx) > gamma;
    fprintf(['Signal %d: Rs = %.0f kBd, estimated fc = %.0f kHz, ' ...
            'T/gamma = %.2f, detected = %s\n'],idx,Rs(idx)/1e3, ...
            peak_fc(idx)/1e3,peak_T(idx)/gamma,string(is_detected(idx)));
end
assert(all(is_detected), 'At least one transmitted signal was not
    detected.');
```

```

assert(all(peak_fc == fc), ...
    'At least one detected carrier does not match its transmitted
    carrier. ');

% Plot the statistic, threshold, and the decisions using shared
thesis style.
[fig, colors] = thesis_figure();
statistic_plot = waterfall(fc_array/1e6,a_array/1e6,T);
statistic_plot.EdgeColor = colors.blue;
statistic_plot.FaceColor = 'none';
hold on
threshold_plot = waterfall(fc_array/1e6,a_array/1e6,threshold);
threshold_plot.EdgeColor = colors.orange;
threshold_plot.FaceColor = 'none';
for idx = 1:numel(Rs)
    if is_detected(idx)
        marker_color = colors.green;
    else
        marker_color = colors.orange;
    end
    scatter3(peak_fc(idx)/1e6,Rs(idx)/1e6,peak_T(idx),50, ...
        marker_color,'filled','MarkerEdgeColor',colors.gray, ...
        'HandleVisibility','off');
end
view(225,20)
title('Streaming SCA Detection of Two M-QAM-SRRC Signals')
xlabel('$f_c$ (MHz)')
ylabel('$R_s$ (MBd)')
zlabel('$T_y(v,k)=|\widehat{C}_y(\alpha_v,f_k)|^2$')
legend([statistic_plot,threshold_plot], ...
    {'$T_y(v,k)$','$\gamma=\beta\widehat{P}_y^2$'}, ...
    'Location','northeast')
thesis_export(fig,'qam_srrc_rs');

```

APPENDIX B

ACRONYMS

ADC	analog-to-digital converter	CFE	center frequency estimator
AF	autocorrelation function	CNR	carrier-to-noise ratio
AGC	automatic gain control	CPU	central processing unit
AWGN	additive white Gaussian noise	CR	cognitive radio
BPSK	binary phase-shift keying	CSD	cyclic spectral density
BRAM	block random-access memory	CSP	cyclostationary signal processing
CAC	cyclic autocorrelation	CubeSat	cube satellite
CAF	cyclic autocorrelation function	DC	direct current
CF	center frequency	DFT	discrete Fourier transform
CFD	cyclostationary feature detector	DPSK	differential phase-shift keying
		DSP	digital signal processor

DSSS	direct-sequence spread spectrum	GPU	graphics processing unit
DVB-S2	digital video broadcasting– satellite–second generation	HDL	hardware description language
ESD	energy spectral density	HSDPA	high-speed downlink packet access
FAM	FFT accumulation method	IID	independent and identically
FCW	frequency control word		distributed
FF	flip-flop	IO	input/output
FFT	fast Fourier transform	IQ	in-phase and quadrature
FIR	finite impulse response	LPF	low-pass filter
FPGA	field- programmable gate array	LSB	least significant bit
		LUT	lookup table
FSK	frequency-shift keying	LUTRAM	lookup-table random-access memory
FSM	frequency- smoothing method	MATLAB	matrix laboratory

MSK	minimum-shift keying	SOC	second-order cyclostationary
NCO	numerically controlled oscillator	SOS	second-order stationary
OOK	on-off keying	SR	symbol rate
PDF	probability density function	SRE	symbol rate estimator
PM	pulse modulation	SRRC	square-root-raised- cosine
PSD	power spectral density	SSCA	strip spectral correlation analyzer
PSF	pulse-shape filter		
PSK	phase-shift keying	SU	secondary user
PU	primary user	SWaP	size, weight, and power
QAM	quadrature amplitude modulation	TSM	time-smoothing method
RF	radio frequency	UMTS	universal mobile telecommunica- tions system
SCA	spectral correlation analyzer	VHDL	VHSIC hardware description language
SNR	signal-to-noise ratio		

VHSIC	very high speed integrated circuit		for microwave access
WSC	wide-sense cyclostationary	WSS	wide-sense stationary
WiMAX	worldwide interoperability		